National Highway Authority of India (NHAI)

(Ministry of Road Transport and Highways)
(Government of India)

PROJECT

Rehabilitation and Upgradation of Existing 2-Lane to 4-Lane standards from end of Kamakhyanagar Bypass - Duburi Section (Ch. 336+900 - Ch. 388+000) of NH 200 (New NH 53) in the State of Odisha on EPC Mode

General Methods & Arrangements for Design and Construction

Revision: R0

(April 2019)

Client	Authority Engineer	Contractor
National Highway Authority of India (NHAI)		G R Infraprojects Limited, Udaipur

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General Methods & Arrangements for Design and Construction

1. Introduction

Table 1- Brief details of the Project

S. No.	Particulars	Details
1.	Project	Rehabilitation and Upgradation of Existing 2-Lane to 4-Lane standards from end of Kamakhyanagar Bypass - Duburi Section (Ch. 336+900 - Ch. 388+000) of NH 200 (New NH 53) in the State of Odisha on EPC Mode
2.	Client	National Highway Authority of India (NHAI)
3.	Authority Engineer	
4.	EPC Contractor	G R Infraprojects Limited, Udaipur (GRIL)
5.	Date of LOA Issued	31-01.2018
6.	Date of Agreement	28-12-2018
7.	Date of Commencement / Appointed Date	23-02-2019
8.	Bid Project Cost	479.27 Cr.
9.	Construction Period	910 Days
10.	Total Length of the Project	51.1 Km
11.	Length of Service/Slip Road	28.71 Km
12.	ROBs	Nil
13.	Grade Separated Structure/ Flyover	Nil
14.	Major Bridge	Reconstruction-01 Nos,Widening-01 Nos
15.	VUPs / CUPs / PUPs	06 Nos VUP
16.	Minor Bridges	New -03 Nos, Reconstruction-06 Nos , Repair/widening-02 Nos.
17.	Interchanges	Nil
18.	Box Culverts	Reconstruction- 72 Nos,New - 10 Nos ,Partial Construction :- 16 Nos.

19.	Pipe Culverts	Widening – 11 Nos, Partial Reconstruction-11 Nos.
20.	Slab Culverts	New Culvert – 06 Nos.
21.	Toll Plaza	
22.	Truck Lay Bye	01 No.
23.	Bus-Bays	46 Nos.
24.	Intersections	Minor- 75 Nos , Major – 02 Nos.

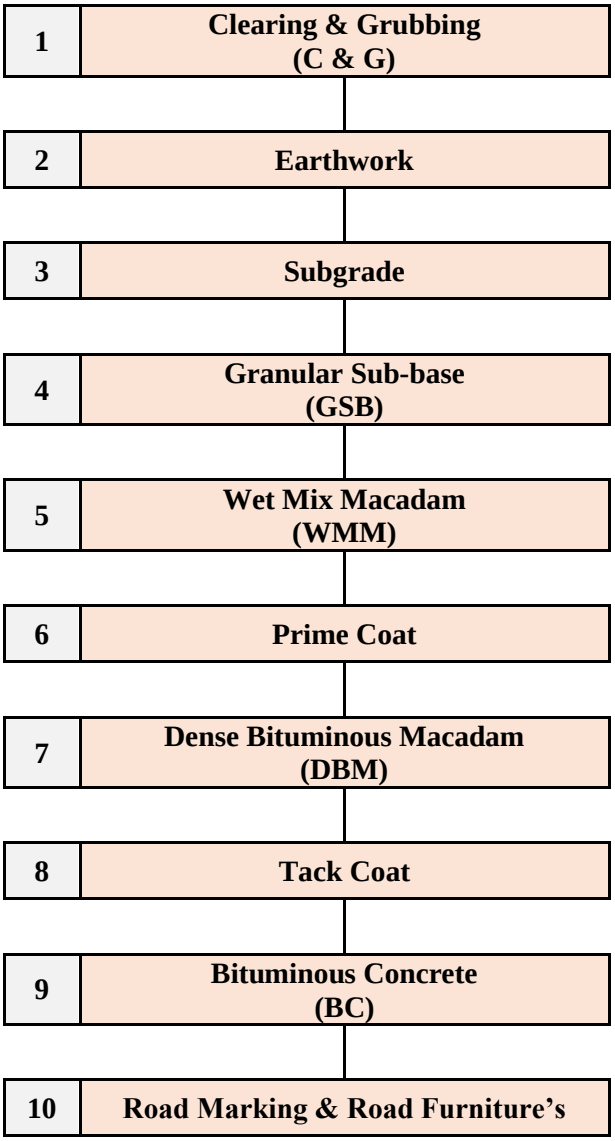
2. Method Statements

The method statement is divided into two sections:

1. Section 1: Road Works which includes Clearing & Grubbing, Survey, Earthwork, GSB, WMM and Bituminous Works, Road Markings, Longitudinal Drainage and Road Appurtenances.
2. Section 2: Structure works which includes the Bridge, ROBs, Grade Separators, and Cross Drainage, Structures & Protection works.

2. 1. Section 1: Road Works

2.1.1. Flow Chart for Road Works



2.1.2. Methodology for Survey and Setting Out

2.1.2.1. Scope of Work

1. The scope of work includes traversing from GPS to GPS, marking of proposed Centre Line and establishment of TBM/BM as per given survey data and their periodic checking as per requirement.
2. Recording of OGL/NGL (Original Ground Level / Natural Ground Level) on grid points at specified intervals.

2.1.2.2. Reference Documents

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.2.3. Instruments / Tools

1. Total Station with all accessories : 1 no.
2. Auto Levels with levelling staff etc. : 1 no.
3. Measuring Tapes : 1 no.

Mason thread, nails, hammer, spade, paint, lime powder etc.

2.1.2.4. Setting Out

1. For all activities of setting out like traversing, marking of proposed Centre Line and TMB checking RFIs will be submitted well in time.
2. The working Bench Mark with reference Bench Mark shall be established and at or near structures.
3. The size to TBM pillars will be 200 x 200 x 700 mm fixed, Precast concrete pillars will be fixed on ground at nearly 250 m interval, their respective values will be fixed by levelling. TBM no. will be written along with their respective R.L. on top or on face of the pillar.
4. The centreline of the carriageway shall be established with Total Station according to the Good For Construction (GFC) drawing data.
5. The proposed centreline points will be marked at every 10 m interval in straight portion and at 5 m intervals at curved portion. For setting out of centre line, control pillar shall be established @ 4 nos per km.
6. Reference pillar 100 x 100 x 500 mm at 50m intervals and at suitable offset will be fixed either side duly marked with chainage on it.

2.1.2.5. Recording of OGL

After fixing of proposed Centre line and benchmarks for recording of natural ground levels (NGL) the grid points will be marked with the measuring tape etc. The staff reading will be recorded on marked grid points. All staff readings (BS, IS & FS) will be recorded in level book and levels will be reduced in reference with the respective TBMs. NGL/OGL recording will be carried out jointly.

2.1.2.6. Checking

The periodic checking for TBM/BM shall be done quarterly with the accuracy for C/L the traversing will be carried out from GPS to GPS.

2.1.2.7. Rectification of Works

Any discrepancies / damage / missing found during checking will be rectified as per instruction of Engineer.

2.1.2.8. Approval

In the beginning and after a periodic check of TBM/BM lists will be prepared jointly, which will be referred for future work.

2.1.3. Methodology for Clearing and Grubbing (C & G)

2.1.3.1. Scope of Work

The work shall consist of cutting, excavation, removing and disposing of all materials such as trees of girth up to 300 mm, bushes, shrubs, stumps, roots, grass weeds, rubbish etc. and top soil up to 150mm declared unsuitable by the Authority Engineer (AE), including draining out stagnant water if any from the area of road land, drain, cross drainage structure and other area as specified in the drawing or instructed by AE. It shall include necessary excavation & back filling of the pits, by suitable soil, resulting from uprooting of trees & stumps and making the surface levelled by suitable equipment and compacted by the power roller to required compaction as per Clause 305.3.4 of the Technical specification. The work also includes handling, salvaging and disposal of cleared material. Clearing and grubbing shall be performed in advance of earthwork operation and in accordance with requirement of this specification.

2.1.3.2. Reference

Reference Documents:

1. Contract Agreement
2. IRC SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)

2.1.3.3. Preservation of Property / Amenities

Suitable safeguards shall be ensured to protect the road side amenities, erosion & water pollution by the exposed surface of slopes & cut formed during the clearing & grubbing operations, shall be prevented through the use of berms, dykes, sediment basins, slope drains, grasses or other suitable devices as per Clause 201.2 of MORTH specifications.

2.1.3.4. Equipment's

The following equipment's shall be used to carry out the Clearing & Grubbing operation.

1. Grader
2. Excavator / J.C.B
3. Tractor Trolley
4. Rollers
5. Water tankers
6. Survey Kit / Instrument
7. Dozer

2.1.3.5. Method of Operation

1. The toe line shall be established, with reference to the alignment based on the levels taken earlier. The area for clearing & grubbing shall be demarcated, covering areas of road embankment, drains, cross drainage structures and such other areas as specified in relevant approved drawings.

2. Trees up to 300 mm in girth and stumps and roots falling within the construction zone shall be removed by the suitable equipment's or by manual labours as per site requirements.
3. The Grader / Excavator / Dozer shall be deployed to remove all shrubs, bushes, vegetation, grass, weeds, rubbish and top organic soil up to 150 mm in thickness, which is unsuitable for incorporation in the permanent work.
4. The above machines & tools shall be deployed to carry out the Clearing & Grubbing operation and the bushes, stumps falling within proposed construction shall be cut to the depth below ground level that in no case these fall within 500 mm of the sub grade. Also all vegetation and deleterious matters unsuitable for incorporation in the embankment & sub -grade shall be also removed between fill lines.
5. All branches of the trees extending above the highway shall be trimmed.
6. All excavation below the general ground level done to remove the trees stumps shall be filled with suitable approved material and compacted to the density as required by the contract specification.
7. The existing soil of natural ground shall be tested for its suitability. In the event of the material non-conforming to the MORTH specifications, it shall be removed and replaced with the approved material & compacted with Vibratory Roller to achieve the required density of 95%.
8. OGL will be taken after clearing, grubbing and compaction.

2.1.3.6. Disposal of the Materials

1. Materials collected from Clearing and Grubbing is disposed-off to the suitable places.
2. Stacking of cleared stumps, branches & boulders shall be done neatly at the appropriate location up to a lead of 1000M.

2.1.3.7. Work Safety

Safety aids shall be mobilized in required number to take care of the workers & of the adjoining road users.

1. All locations adjacent to the existing road where clearing & grubbing is progressing shall be provided with visible safety & road signs.
2. Minimum of two flagmen with reflective vests holding a red flag each shall be deployed to control the traffic wherever required during clearing & grubbing operations.
3. No machines or equipment's shall be permitted to remain on the existing road or Shoulder unattended unless it is properly protected &secured in a safe manner.
4. The Safety Officer frequently patrols at construction zones of project highway to ensure the safety of equipment, workers & road users and safety signs are operational at all time.

2.1.3.8. Environmental Safety

1. Deleterious matters detrimental to the environment shall be disposed-off as directed by AE at the designated locations provided by the Authority.
2. Vehicles tracks shall be kept moist to prevent flying of dust particles with the vehicle movement.

2.1.4. Methodology for Excavation

2.1.4.1. Scope of Work

The work shall consist of excavation, removal and disposal of materials necessary for the construction of roadway, side drains and waterways in accordance with clause 301 of MORTH specifications.

2.1.4.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.4.3. Setting Out

After the site has been cleared as per clause 201, the limits of excavation shall be set out true to lines, curves, slopes, grades and sections as shown in the drawings.

2.1.4.4. Classification

All materials involved in excavation shall be classified as following:

1. Soil: Removal of gravel or any other modular material having dimension in any one direction not exceeding 75mm be deemed to be covered under this category.
2. Ordinary Rock: Boulders which do not require blasting found lying loose in the surface or embedded in river bed, soil, talus, slope wash and terrace material of dissimilar origin are included in this category. Ex. Varieties of limestone and sandstone, lime concrete, stone masonry and brickwork.
3. Hard Rock: Any rock or cement concrete for the excavation of which the use of mechanical plant and/or blasting is required.

   Hard rock using controlled blasting is carried out in locations where built-up area, huts are located within 200m of blast site. Blasting is prohibited in areas where people are living within 20m of blast site. In such cases the excavation has to be carried out by Chiselling, Wedging or any other agreed method.

2.1.4.5. Method of Operation

1. Rock Excavation: Rock, when encountered in road excavation, shall be removed up to the formation level. If unsuitable materials are encountered at formation level, these shall be excavated to the extent of 500 mm below the formation level. Where excavations is done to levels lower than those specified, the excavation shall be made as per clause 301.3.3. If blasting is required, the same shall be carried out as per clause 302 with suitable precautions.

2.1.4.6. Backfilling

Backfilling of masonry/concrete pipe or drain excavation shall be done with approved material in such a way as not to cause differential settlement.

2.1.4.7. Preservation of Property

Suitable safeguards shall be ensured to protect and preserve of any or all existing roadside trees, drains, sewers, sub surface drains, pipes, conduits or any other structure under or above ground as per clause 301 of MORTH.

2.1.4.8. Finishing operations

Finishing operations shall include the work of properly shaping and dressing all excavated surfaces. When completed, no point on the slopes shall vary more than 150 mm measured at right angles to the slope except in case of rock excavation. The depth of top soil shall be sufficient to sustain plant growth with thickness between 75 mm to 100 mm as mentioned in clause 301.7 of MORTH specifications.

2.1.5. Methodology for Embankment Construction

2.1.5.1. Scope of Work

The work shall consist of construction of embankment with approved and specified materials obtained from approved borrow areas or suitable material obtained from roadway excavation and drain excavation and in accordance with clause 305 of MORTH specifications.

2.1.5.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.5.3. Setting Out

After completion of site clearance, the limits of embankment shall be marked by fixing pegs on both sides at regular intervals. The chainage boards and working bench mark shall be set outside the limits of construction areas.

2.1.5.4. Selection of Material and Borrow areas

1. Material: The material used in embankment shall be soil, moorum, gravel, reclaimed material from pavement, fly ash, pond ash, a mixture of these or any other material approved by the AE. It shall be free of logs, stumps, roots, rubbish & any other material likely to deteriorate or affect the stability of the embankment. The material for embankment shall be obtained from the approved source with preference to the material becoming available from nearby roadway excavation or any other excavation under the contract. The material requirements shall be in accordance with clause 305.2 of MORTH specifications.
2. Borrow Material: The Borrow Material shall comply the Clause 305.2.2.2 of the MORTH specifications.

2.1.5.5. Equipment's

The following equipment's shall be used for the Embankment Construction are:

1. Hydraulic Excavator
2. Dozer
3. Dumper (Tipper Trucks)
4. Motor Grader
5. Vibratory Roller
6. Water Tanker
7. Tractor Trolleys
8. Survey kit / instrument

2.1.5.6. Method of Operation:

1. After completion of site clearance, the limits of embankment shall be marked by fixing pegs at regular intervals before commencing the earthwork. Construction toe line will be more than the design toe line to ensure proper compaction on edges.
2. The original ground shall be levelled to facilitate placement of first layer of embankment, scarified, mixed with water and then compacted by rolling so as to achieve minimum dry density as specified in the Table 300-1 of MORTH specifications. Unsuitable material occurring in embankment foundation shall be removed and replaced by approved materials laid in layers and suitably compacted to the required and specified degree of compaction. Excavation at approved borrow areas shall be carried out with excavator and transportation of excavated material shall be done by dumpers / tractor trolley.
3. Natural ground soil shall be tested for its suitability. If found unsuitable it shall be replaced with the suitable material.
4. M.D.D. of natural ground soil shall be determined, and natural ground shall be tested for dry density, if found less than 95% of M.D.D., the original ground shall be scarified mixed with water and compacted with vibratory roller to achieve minimum specified dry density.
5. The embankment material shall then be spread in layers up to 250 mm (when compacted using vibratory roller / compactor) or up to 200 mm (when compacted using 80-100 kN static roller) thicknesses over the entire area with the dozer and finished by a motor grader. Where the existing embankment to be widened continuous horizontal benches shall be cut out into the old slope to ensure adequate bond. Where water is required to be added the same shall be sprinkled from a water tanker fitted with sprinkler, uniformly on the surface but without flooding and shall be mixed thoroughly in soil by harrowing until uniform moisture content is obtained throughout the depth of the layer. If the material delivered to the roadbed is too wet, it shall be dried by aeration and exposure to the sun till the moisture come down at/near the OMC level.
6. The compaction shall be done with the help of vibratory roller or 8 - 10 tonne static roller. Compaction trial shall be carried out on a suitable stretch to determine the no. of passes required for particular type of soil, dumped for embankment, to achieve densities specified in the Table 300-2 of MORTH specifications. Based on the trial actual compaction will be carried out with required no. of passes of vibratory roller until the specified density is achieved. Rolling shall progress parallel to the centreline of the road uniformly overlapping each preceding track by one-third width. Rolling shall be continued till the specified density is achieved as per Table 300-2. In case specified compaction is not achieved the material in the soft area shall be removed and replaced with approved material. Densities shall be tested by sand replacement method.
7. Subsequent layers shall be placed only after the finished layer has been tested as per the clause 903.2.2 and accepted. The top levels of the embankment shall be constructed with the approved drawing keeping the variation within tolerance limit as per Table 900-1.

2.1.6. Methodology for Embankment with Fly-ash

2.1.6.1. Scope

The scope of work includes the details regarding the construction of the embankment with fly-ash.

2.1.6.2. Reference Documents

1. Contract Agreement
2. IRC SP: 84-2014
3. IRC SP: 58-2001
4. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)

2.1.6.3. Setting Out

After completion of site clearance, the limits of embankment shall be marked by fixing pegs on both sides at regular intervals. The chainage boards and working bench mark shall be set outside the limits of construction areas.

2.1.6.4. Material

The materials to be used in embankment construction should be characterised to determine their physical and engineering properties. The fly-ash conforming to specifications as per IRC SP: 58-2001 shall be used.

2.1.6.5. Clearing and Grubbing

This work consists of cutting, removing and disposal of trees, bushes, shrubs, roots, grass, rubbish, etc., from the alignment and within the area of road land which will accommodate road embankment, drains, and such other areas as specified on the drawings. During clearing and grubbing, the contractor shall take adequate precautions against soil erosion, water pollution, etc. All trees, stumps, etc., falling within fill area should be cut to atleast 500 mm below ground level and pits shall be filled with suitable material and compacted thoroughly so as to make the surface at these points conform to the surrounding area.

2.1.6.6. Stripping and Storing of Top Soil

When constructing embankment using fly ash, the top soil from all areas to be covered by the embankment foundation should be stripped to specified depth not exceeding 150 mm. Top soil should not be unnecessarily trafficked either before stripping or when in stockpiles. Also, these shall not be surcharged or otherwise loaded and multiple handling should be kept to minimum.

2.1.6.7. Setting Out

After the site has been cleared, the limits of embankment should be set out true to lines, curves, slopes, grades and sections as shown on the drawings. The limits of the embankment should be marked by fixing batter pegs on both sides at regular intervals as guides before commencing the construction. The embankment should be built sufficiently wider than the design dimensions so that surplus material may be trimmed, ensuring that the remaining material is of the desired density and in position specified, and conforms to the specified slopes. Bench marks and other stakes should be maintained as long as in the opinion of the engineer, they are required for the work.

2.1.6.8. Compacting the Ground supporting Embankment

Where necessary, the original ground should be levelled, scarified, mixed with water and then compacted by rolling so as to achieve minimum 97 percent of MDD determined as per IS:2720 (Part 8) for the foundation soil.

2.1.6.9. Handling and Transportation of Fly-Ash

Fly ash is typically delivered to the site in covered dumper truck to minimize loss of moisture and dusting. Fly ash generally contains moisture to prevent dusting.

2.1.6.10. Spreading and Compaction

Intermediate soil layers are often provided in the fly ash embankment for ease of construction, to facilitate compaction of ash and to provide adequate confinement. Such layers minimise liquefaction potential also. Embankment with intermediate soil layers can be adopted in case height of the embankment is more than 3 m. The compacted thickness of intermediate soil layers shall not be less than 200 mm. One or more layers shall be constructed depending upon the design requirements. The vertical distance between such layers may vary from 1 .5 to 3 m. The top 0.5 m of embankment should be constructed using selected earth to form the subgrade for the road pavement. Typical cross-sections of fly ash embankment with and without intermediate soil layers are shown in Fig. 1 below.

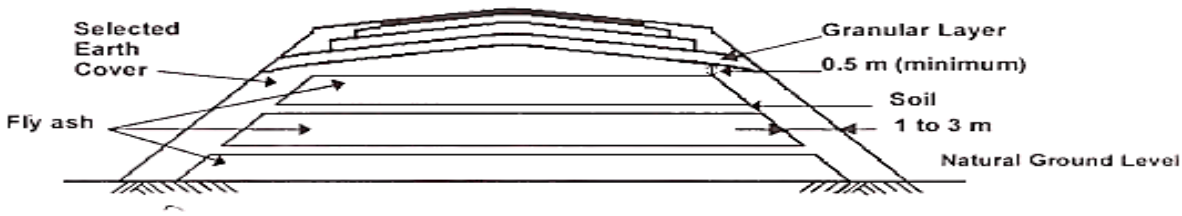


Fig. 1. Typical Cross-Section of Embankment with Alternate Layer of Fly Ash and Soil

The side soil cover of required width shall be provided along with the core and mechanically compacted as the embankment progresses upwards. The addition of side cover subsequent to the construction of the core is prohibited. The fill material should be spread by mechanical means, finished by motor grader. Smaller vibratory rollers with dead weights of 10 to 15 kN perform well on loose layer thickness of the order of 100-150 mm. Medium weight vibratory rollers with dead weights in the range 60-100 kN, provide satisfactory compaction for loose layer thickness of about 250 mm. When vibratory roller of dead weight 80-100 kN are used, loose layer thickness up to 400 mm can be adopted. When compaction is earned out using only static roller of 80- 100 kN weight, loose layer thickness shall not exceed 200 mm. The cover soil and fly-ash should be laid simultaneously before compaction, to ensure confinement of fly-ash. Clods or hard lumps in cover soil shall be broken to have a maximum size of 50 mm.

Moisture content of the fill material shall be checked at the site prior to commencement of compaction. Moisture content of fly-ash laid for compaction shall normally vary from OMC± 2% (determined as per IS: 2720 (Part 8)). The moisture content limits can be varied depending on the weather conditions, by the Engineer-in-charge, provided specified compaction is achieved as revealed through actual site trials and there is no dust problem. At moisture contents higher than the appropriate range, fly ash may liquefy and would be difficult to handle and compact. Moisture content of cover soil shall be maintained at its OMC. Where water is required to be added to the fill material, it shall be sprinkled from a water tanker fitted with a sprinkler capable of applying water uniformly without any flooding. The water shall be mixed thoroughly by blading, discing or harrowing or by suitable means until uniform moisture content is obtained throughout the depth of the layer. If the material delivered to the construction site is too wet, it shall be dried by aeration and exposure to sun, till the moisture content is acceptable for compaction.

Fly ash can be compacted using vibratory or static rollers. Each layer of fly-ash shall be thoroughly compacted to the specified density. The construction of fly ash core and earth cover on the sides should proceed simultaneously. Each compacted layer shall be finished parallel to

the final cross-section of the embankment. The minimum dry density after compaction as percentage of MDD IS:2720 (Part 8) shall be 95% except when used in bridge abutments - for embankment width equal to 1 .5 times the height of the embankment where 100% compaction shall be required.

Fly ash embankments should be benched at 4 to 6 m vertical intervals to drain surface water run-off to the ends of the embankment, rather than allowing full volume of the run-off to travel down the face of the embankment to the toe. Run-off from pavement surfaces should be collected and discharged into proper drainage system.

2.1.6.11. Finishing Operations

Finishing operations shall include the work of shaping and dressing the shoulders/verge/road bed and side slopes to conform to the alignment, levels, cross-sections and dimensions shown on the drawing or as directed by the Engineer subject to the tolerance. Both upper and lower ends of side slopes shall be rounded off to improve appearance and to merge the embankment with the adjacent terrain.

2.1.6.12. Quality Control

Quality of compacted material shall be controlled through periodic checks on the compaction process or the end product, singly or in combination as directed. The end product must conform to the specifications. The number of tests to be conducted and acceptance criteria shall be as outlined in MORTH Specifications for Road and Bridge Works, Section 900.

2.1.7. Methodology for Subgrade Construction

2.1.7.1. Scope of Work

The work shall consist of construction of Subgrade with approved and specified materials obtained from approved borrow areas or suitable material obtained from roadway excavation and drain excavation and in accordance with clause 305 of MORTH specifications.

2.1.7.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.7.3. Setting Out

After completion of site clearance/embankment the limits of sub-grade shall be marked by fixing pegs on both sides at regular intervals. The working bench mark shall be set outside the limits of construction areas.

2.1.7.4. Selection of Material and Borrow Areas

1. Material: The material used in Sub-grade shall be soil, moorum, gravel, reclaimed material from pavement, fly ash, pond ash, a mixture of these or any other material approved by the AE. It shall be free of logs, stumps, roots, rubbish & any other material likely to deteriorate or affect the stability of subgrade. The material for subgrade shall be obtained from the approved source with preference to the material becoming available from nearby roadway excavation or any other excavation under the contract. The material requirements shall be in accordance with clause 305.2 of MORTH specifications.

2. Borrow Material: Samples shall be collected from the identified borrow area & tested as per IS 2720. On confirmation of the suitability material the area shall be properly demarcated.

2.1.7.5. Equipment

The following equipment shall be carried out for the subgrade construction. If desired the Contractor shall demonstrate the efficacy of the type of equipment to be used, before commencement of work.

1. Hydraulic Excavator
2. Dozer
3. Dumper (Tipper Trucks)
4. Motor Grader
5. Vibratory Roller
6. Water Tanker
7. Tractor Trolleys
8. Survey Kit

2.1.7.6. Method of Operation

1. After completion of embankment, the limits of subgrade shall be marked by fixing pegs at regular intervals before commencing the earthwork. Construction toe line will be more than the design toe line to ensure proper compaction on edges.
2. The embankment top shall be levelled to facilitate placement of first layer of subgrade, scarified, mixed with water and then compacted by rolling so as to achieve minimum dry density as specified in Table 300-1 of MORTH specifications. Subgrade treatment specified for subgrade shall be carried out as per approved design and drawing. Unsuitable material occurring in subgrade foundation shall be removed as per direction and approval of AE and replaced by approved materials laid in layers and suitably compacted to the required and specified degree of compaction. Excavation at approved borrow areas shall be carried out with excavator and transportation of excavated material shall be done by dumpers.
3. M.D.D. of borrow area soil as determined in lab, and when the same be laid for subgrade preparation and tested for dry density should not have relative compaction less than 97% of M.D.D. (as specified in MORTH specifications Table 300-2), if found less, the layer shall be scarified mixed with water and compacted to achieve minimum specified dry density.
4. The subgrade material shall then be spread in layers not exceeding 250 mm (when compacted using vibratory roller / compactor) or up to 200 mm (when compacted using 80-100 kN static roller) thicknesses over the entire area with the dozer and finished by a motor grader. When the existing subgrade to be widened continuous horizontal benches shall be cut out into the old slope to ensure adequate bond. Where water is required to be added the same shall be sprinkled from a water tanker fitted with sprinkler, uniformly on the surface but without flooding and shall be mixed thoroughly in soil by blending or harrowing until uniform moisture content is obtained throughout the depth of the layer. If the material delivered to the roadbed is too wet, it shall be dried by aeration and exposure to the sun till the moisture content is accepted for compaction.
5. The compaction shall be done with the help of vibratory roller or 8 - 10 tonne static roller. Compaction trial shall be carried out on a suitable stretch to determine the no. of passes required for particular type of soil, dumped for subgrade, to achieve densities specified in the Table 300-2 of MORTH specifications. Based on the trial actual compaction will be carried out with required no. of passes of vibratory roller or static roller until the specified density is achieved. Rolling shall progress parallel to the

centreline of the road uniformly overlapping each preceding track by one-third width. Rolling shall be continued till the specified density is achieved as per Table 300-2 of MORTH specifications. In case specific compaction is not achieved the material in the soft area shall be removed and replaced with approved material. Densities shall be tested by sand replacement method.

6. Subsequent layers shall be placed only after the finished layer has been tested as per the clause 903.2.2 and accepted. The top levels of the subgrade shall be checked with reference to the longitudinal and cross profile of the road as per drawing to keep the variation within tolerance limit as per table 900-1.
7. The Safety Officer shall make frequent patrols along the highway to ensure that the safety equipment and signs are operational at all time.

2.1.8. Methodology for Granular Sub-base (GSB)

2.1.8.1. Scope of Work

This work shall consist of laying and compacting well-graded material on prepared subgrade in accordance with the requirements of MORTH specifications. The material shall be laid in one / two layer as necessary according to lines, grades and cross-sections shown on the drawings or as directed by the AE.

2.1.8.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. IRC: 37-2012
4. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
5. Relevant Drawings

2.1.8.3. Material

The material to be used for the work shall be natural sand, moorum, gravel, crushed stone, crushed slag, or combination thereof depending upon the grading required. Materials like brick metal, kankar and crushed concrete may be allowed only with the specific approval of the AE. The material shall be free from organic or other deleterious constituents and shall conform to the quality standards as prescribed in the Table 400-1 of MORTH specifications and reproduced below.

IS Sieve Designation	Percent by Weight Passing the IS Sieve					
	Grading 1	Grading 2	Grading 3	Grading 4	Grading 5	Grading 6
75.0 mm	100	-	100	-	100	-
53.0 mm	80-100	100	-	-	80-100	100
26.5 mm	55-90	70-100	55-75	-	55-90	75-100
9.50 mm	35-65	50-80	-	50-80	35-65	55-75
4.75 mm	25-55	40-65	10-30	-	25-50	30-55
2.36 mm	20-40	30-50	-	-	10-20	10-25
0.425 mm	Oct-15	Oct-15	-	15-35	0-5	0-8
0.075 mm	< 5	< 5	< 5	< 5	-	0-3

Aggregate Impact Value (AIV) / Wet AIV	Max 40
Liquid Limit	Max 25
Plasticity Index	Max 6
CBR Value	Min 30%

If the water absorption of aggregates determined as per IS: 2386 (Part 3) is greater than 2%, the aggregates shall be tested for Wet Aggregate Impact Value (Wet AIV) as per IS: 5640.

2.1.8.4. Construction Operations

1. Preparation of Subgrade: Immediately prior to the laying of GSB, the subgrade already finished to Clause 301 or 305 as applicable shall be prepared by removing all vegetation and other extraneous matter, lightly sprinkled with water and rolled with two light passes of 80-100 kN smooth wheeled roller if necessary.
2. The sub-base material of the specified grading and water shall be mixed mechanically be a suitable mixer equipped with provision for controlled addition of water and mechanically mixing.
3. Spreading & Compacting: The sub-base material of grading specified in MORTH specifications shall be spread on the prepared subgrade with the help of a motor grader of adequate capacity, its blade having hydraulic controls suitable for initial adjustment and for maintaining the required slope and grade during the operation or other means as approved by the AE. The thickness of each compacted layer shall not be less than 100mm.
4. The moisture content of the mix shall be checked in accordance with IS 2720 (Part 2) and suitably adjusted so that at the time of compaction it is from 1 to 2% below optimum moisture content (OMC).
5. Immediately after spreading the mix, rolling shall be done by a smooth wheeled roller of 80-100 kN weight for a compacted thickness less than 100 mm and with a vibratory roller of 80-100kN static weight for the thickness up to 200 mm. Rolling shall commence at the lower edge and proceed towards the upper edge longitudinally for portions having unidirectional cross fall or on super elevation. For carriageway having cross fall on both sides, rolling shall commence at the edges and progress towards the crown.
6. Rolling shall be continued till the density achieved is at least 98 percent of the maximum dry density for the material determined as per IS: 2720 (Part 8). The surface of any layer of material on completion of compaction shall be well closed, free from movement under compaction equipment and from compaction planes, ridges, cracks or loose material. All loose, segregated or otherwise defective areas shall be made good to the full thickness of layer and re-compacted.

2.1.8.5. Surface Finish and Quality Control of Work

The surface finish of construction shall conform to the requirements of Clause 902. Control on the quality of materials and works shall be exercised by the AE in accordance with Contract Agreement and MORTH specifications Section 900.

2.1.9. Methodology for Wet Mix Macadam (WMM)

2.1.9.1. Scope

This work shall consist of laying and compacting clean, crushed, grade aggregate and granular material, premixed with water, to a dense mass on a prepared GSB. The material shall be laid in one or more layers as necessary to lines, grades and cross- sections shown on the approved drawings or as directed by the AE.

The thickness of a single compacted WMM layer shall not be less than 75 mm. When vibrating or other approved types of compacting equipment are used, the compacted depth of a single layer of the WMM may be up to 200 mm.

2.1.9.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. IRC: 37-2012
4. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
5. Relevant Drawings

2.1.9.3. Materials

1. Aggregates:
   a. Physical Requirements: Coarse aggregates shall be crushed stone. If crushed gravel / shingle is used, not less than 90 percent by weight of the gravel / shingle pieces retained on 4.75 mm sieve shall have at least two fractured faces. The aggregates shall conform to the physical requirements set forth in Table 400-12 of MORTH specifications.
   b. Grading Requirements: The aggregates shall conform to the grading given in Table 400-13 of MORTH specifications and reproduced below.

IS Sieve Designation, mm	% by weight passing the IS Sieve
53.00	100
45.00	95-100
26.50	-
22.40	60-80
11.20	40-60
4.75	25-40
2.36	15-30
0.600	8-22
0.075	0-5

2.1.9.4. Construction Operations

1. Preparation of Base: The surface of the sub-base to receive the Wet Mix Macadam course shall be prepared to the specified grade and camber and cleaned of dust, dirt and other extraneous material. Any ruts or soft yielding places shall be corrected in an approved manner and rolled until firm surface is obtained.

2. Provision for Lateral Confinement of Aggregates: While constructing WMM, arrangement shall be made for the lateral confinement of wet mix. This shall be done by laying materials in adjoining shoulders along with that of wet mix macadam layer and following the sequence of operations described in Clause 404.3.3.
3. Preparation of Mix: WMM shall be prepared in an approved mixing plant of suitable capacity having provision for controlled addition of water and forced / positive mixing arrangement like pug mill or pan type mixer of concrete batching plant. The plant shall have following features:
   a. Cold aggregates with minimum four bin feeders with variable speed motor
   b. Vibrating screen for removal of oversize aggregates
   c. Conveyor Belt
   d. Controlled system for addition of water
   e. Forced / positive mixing arrangement like pug-mill or pan type
   f. Centralized control panel for sequential operation of various devices and precise process control
   g. Safety devices

Optimum moisture for mixing shall be determined in accordance with IS: 2720 (Part-8) after replacing the aggregate fraction retained on 22.4 mm sieve with material of 4.75 mm to 22.4 mm size. While adding water, due allowance should be made for evaporation losses. However, at the time of compaction, water in the wet mix should not vary from the optimum value by more than agreed limits. The mixed material should be uniformly wet and no segregation should be permitted.

2.1.9.5. Spreading of Mix

Immediately after mixing, the aggregates shall be spread uniformly and evenly upon the prepared sub-base in required quantities. In no case shall these be dumped in heaps directly on the area where these are to be laid nor shall their hauling over a partly completed stretch be permitted.

2.1.9.6. Compaction

After the mix has been laid to the required thickness, grade and cross-fall / camber the same shall be uniformly compacted to the full depth with suitable roller. If the thickness of single compacted layer does not exceed 100 mm, a smooth wheel roller of 80 to 100kN weight may be used. For a compacted single layer up to 200 mm, the compaction shall be done with the help of vibratory roller of minimum static weight of 80 to 100kN with an arrangement of adjusting frequency and amplitude. The speed of the roller shall not exceed 5km/h.

2.1.9.7. Setting & Drying

After final compaction of wet mix macadam course, the road shall be allowed to dry for 24 hours.

2.1.9.8. Opening of Traffic

No vehicular traffic of any kind should be allowed on the finished wet mix macadam surface. Construction traffic may be allowed.

2.1.9.9. Quality Control & Surface Finish

Control on the quality of materials and works shall be exercised by the Engineer in accordance with Section 900. The surface finish of construction shall conform to the requirements of Clause 902.

2.1.9.10. Rectification of Surface Irregularities

Where the surface irregularity of the WMM course exceeds the permissible tolerances or where the course is otherwise defective due to sub-grade soil getting mixed with the aggregates, the full thickness of the layer shall be scarified over the affected area, re-shaped with added premixed material or removed and replaced with fresh premixed material as applicable and re-compacted in accordance with Clause 406.3. The area treated in the aforesaid manner shall not be less than 5 m long and 2 m wide. In no case shall depressions be filled up with unmixed and ungraded material or fines.

2.1.10. Methodology for Prime Coat

2.1.10.1. Scope

The work shall consist of application of single coat of low viscosity liquid bituminous material over prepared WMM surface preparatory to the superimposition of bituminous treatment or mix.

The prime coat shall be applied only on WMM layer over which DBM layers are to be laid.

The priming shall not be carried out if the surface of base layer is wet, or if the weather is foggy, rainy, windy etc. or the temperature in shade is less than 10ºC.

The rate of spray shall be as specified in section 502.2.2 of the MORTH specifications i.e. 0.7 to 1 kg/sq.m.

2.1.10.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.10.3. Materials

The primer used for prime coat shall be cationic bitumen emulsion complying with IS 8887. The particular grade and test certificate of prime coat will be submitted to the AE for approval.

2.1.10.4. Equipment's

1. Priming shall be carried out by using self-propelled bitumen pressure distributor.
2. The distributor shall ensure that the material is sprayed uniformly and at specified rates and temperatures.
3. Air Compressor.
4. Mechanical Broom.

2.1.10.5. Preparation of Base

The base layer surface to be primed shall be swept clean, made free from dust and be made dry. It shall be shaped to the specified grades and levels as per drawings. It shall be free from any ruts, irregularities and segregated materials.

If the surface to be primed is so dry and dusty so as to cause freckling of bituminous material, the same shall be lightly and uniformly sprinkled with water immediately before priming but priming shall be done after no surface water is visible.

2.1.10.6. Application

The prime coat shall be sprayed uniformly over the base surface. The temperature of the primer during application shall be such that the primer could be sprayed effectively through the jets of the spray bar.

2.1.10.7. Curing

After the application of the Prime Coat, the surface shall be allowed to be cured for at least 24 hours so as to allow the penetration of the Priming Material into the Base course. During this curing period no traffic of any kind shall be allowed to ply on the surface.

2.1.10.8. Arrangement of Traffic

No traffic will be allowed on priming surface. Diversion road if required will be provided as required under Clause 112, of the MORTH specifications.

2.1.11. Methodology for Dense Bituminous Macadam (DBM)

2.1.11.1. Scope

This work shall consist of construction in a single or multiple layers of DBM on a previously prepared base or sub base. The thickness of single layer shall be between 50 mm to 100 mm or as per approved design and drawings.

2.1.11.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.11.3. Equipment's

1. Crusher of required capacity for Aggregates/ Crushed Aggregate
2. Hot Mix Plant
3. Paver with sensor arrangement
4. Steel Tandem Rollers with required weight
5. Pneumatic Tandem Roller with required weight
6. Bitumen Sprayer
7. Dumpers
8. Air compressor
9. Pegs for putting levels
10. Broom (Mechanical)
11. Straight edge
12. Thermometers (0 - 250ºC range)
13. Gauge for checking thickness
14. Survey Kit / instrument
15. Measuring tape (metal)
16. Core cutting machine

2.1.11.4. Materials

The source and test report of all material shall be submitted to the AE for approval prior to use in aforesaid work.

1. Bitumen: The type & grade of bitumen shall be as specified in the CA or as per Pavement Design. If material is straight run bitumen it should confirm to IS 73 and in case of modified bitumen it shall be as per IRC SP 53.

2. Coarse Aggregates: The coarse aggregates shall consist of crushed rock, crushed gravel or other hard material retained on 2.36 mm sieve. They shall be clean, hard, durable of cubical shape, free from dust and soft or friable matter, organic or other deleterious substances. Coarse aggregates shall be procured from the approved quarry and crusher. The coarse aggregate shall satisfy the specifications requirement set out in sub-clause 505.2.2 and 505.2.5 of the MORTH specifications i.e. as given below:

Property	Test	Specification	Testing Method
Cleanliness	Grain Size analysis	Max 5% passing 0.075mm Sieve	IS:2386 Part-I
Particle Shape	Flakiness and Elongation Index(Combined)	Max 35%	IS:2386 Part-I
Strength	Los Angeles Abrasion Value or Aggregate Impact Value	Max 35% Max 27%	IS:2386 Part-IV IS:2386 Part-IV
Durability	Soundness either Sodium Sulphate or Magnesium Sulphate	Max 12% Max 18%	IS:2386 Part-V IS:2386 Part-V
Water Absorption	Water Absorption	Max 2%	IS:2386 Part III
Stripping	Coating and Stripping of Bitumen Aggregate	Min. Retained Coating 95%	IS:6241
Water Sensitivity	Retained Tensile Strength	Min. 80%	AASTHO 283

Note: If the minimum retained tensile test strength falls below 80%, anti-stripping agent shall be used.

3. Fine Aggregates: Fine aggregates shall be either crushed or naturally occurring mineral material, or a combination of the two, passing the 2.36 mm sieve and retained on the 75 micron sieve. These shall be clean, hard, durable, dry and free from dust, and soft or friable matter, organic or other deleterious matter. The fine aggregate shall have a sand equivalent value of not less than 50 when tested in accordance with the requirement of IS: 2720 (part 37). The plasticity index of the fraction passing the 0.425 mm sieve shall not exceed 4, when tested in accordance with IS: 2720 (Part 5).

4. Filler: This shall conform to Clause 505.2.4 of MORTH specifications. Filler shall consist of finely divided mineral matter such as rock dust, hydrated lime or cement. The filler shall be graded within the limits indicated in Table 500-9 of MORTH specifications.

IS sieve (mm)	Cumulative Percent Passing by weight of Total Aggregate
0.6	100
0.3	95-100
0.075	85-100

5. Aggregate Grading and Binder Content: When tested in accordance with IS: 2386 Part I (wet sieve method), the combined grading of the coarse and fine aggregate and added filler for the particular mixture shall fall within the limits shown in Table 500-10 of MORTH specifications. Bitumen content shall be determined in accordance with clause 505.3 of MORTH specifications.

2.1.11.5. Mix Design

1. Requirement of Mix: Apart from conformity with the grading and quality requirement for individual ingredients, the mixture shall meet the requirements set out in table 500-11 of the MORTH specifications.
2. Mix Design: The binder content shall be optimized to achieve the requirement of the mix set out in table 500-11. The Marshall method for determining the optimum binder content shall be adopted as described in MS-2 Asphalt Institute Manual. The Fines to Bitumen (F/B) ratio by weight of total mix shall range from 0.6 to 1.2.
3. Job Mix Formula: The procedure for formulating the job mix formula shall be generally as specified in clause 505.3.3 of MORTH specifications. All relevant testing and Mix design shall be conducted jointly along with AE. The test results obtained thereafter shall be submitted to the AE for approval and trial stretch will be laid based on the submitted JMF.
4. Plant Trials: After completion of JMF at lab, plant trial will be carried out to establish that the plant can be set up to produce a uniform mix conforming to the approved job mix formula. The permissible variation in job mix formula will be as per the limits specified in MORTH specifications Table500-13 and reproduced below.

Description	Base/Binder Course
Aggregate Passing 19mm Sieve or larger	±8%
Aggregate passing 13.2mm, 9.5mm sieve	±7%
Aggregate passing 4.75mm	±6%
Aggregate passing 2.36mm, 1.18mm, 0.6mm	±5%
Aggregate passing 0.3mm, 0.15mm	±4%
Aggregate passing 0.075mm	±2%
Binder content	±0.3%
Mixing temperature	±10 ºC

2.1.11.6. Construction Operation

1. Weather and Seasonal Limitations: The provisions of clause 501.5.1 MORTH specifications shall be followed.
2. Prime Coat: Where the material on which the DBM is to be laid is other than a bitumen bound layer, a prime coat shall be applied in accordance with clause 502.
3. Preparation and Transportation of Mix: The provisions of clauses 501.3 and 501.4 shall apply. The mixing, laying and rolling temperature shall be as per Table 500-2 of MORTH specifications.
4. Spreading: The provisions of clause 501.5.3 & 501.5.4 of Morth shall apply.
5. Rolling: Compaction shall be carried out in accordance with the provisions of clauses 501.6 and 501.7 The density of the finished paving layer shall be determined by taking

cores, no sooner than 24 hours after laying, or by other approved method. The compacted layers of DBM shall have a minimum field density equal to or more than 92% of the density based on theoretical maximum specific gravity (G_{mm}) obtained on the day of compaction. Rolling shall be continued until the specified density is achieved or there is no movement under the roller.

2.1.11.7. Surface Finish and Quality of work

The surface finish of the completed construction shall conform to the requirement of clause 902. For control of the quality of materials and the work carried out, the provisions of section 900 shall apply.

2.1.12. Methodology for Tack Coat

2.1.12.1. Scope

The work shall consist of application of single coat of low viscosity liquid bituminous material over DBM surface by clause 503 of the MORTH specifications.

The tack coat shall not be carried out if the surface of DBM layer is wet, or if the weather is foggy, rainy, windy etc. or the temperature in shade is less than 10ºC.

The rate of spray shall be as specified in Table 500-5 of the MORTH specifications i.e. 0.2 to 0.3kg/sqm.

2.1.12.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.12.3. Materials

The binder used for tack coat shall be cationic bitumen emulsion (RS 1) complying with IS 8887 or suitable low viscosity paving bitumen of VG 10 grade conforming to IS: 73.

2.1.12.4. Equipment's

1. Tack coat shall be carried out by using self-propelled bitumen pressure sprayer for spraying the material uniformly at a specified rate and temperatures.
2. Air Compressor.
3. Mechanical Broom.

2.1.12.5. Preparation of Base

The DBM layer surface on which tack coat is to be applied shall be swept clean, made free from dust and be made dry. It shall be shaped to the specified grades and levels as per drawings. It shall be free from any ruts, irregularities and segregated materials.

2.1.12.6. Application

The application of tack coat shall be at the rate specified in Table 500-5, and it shall be applied uniformly. If rate of application of Tack Coat is not specified in the contract, then it shall be the rate specified in Table 500-5. No dilution or heating at site of RS1 bitumen emulsion shall be permitted. Paving bitumen if used for tack coat shall be heated to appropriate temperature in bitumen boilers to achieve viscosity less than 2 poise. The normal range of spraying temperature for a bituminous emulsion shall be 20°C to 70°C and for cutback, 50°C to 8O°C.

The method of application of tack coat will depend on the type of equipment to be used, size of nozzles, pressure at the spray bar, and speed or forward movement.

2.1.12.7. Curing

The tack coat shall be left to cure until all the volatiles have evaporated before any subsequent construction is started. No plant or vehicles shall be allowed on the tack coat other than those essential for the construction.

2.1.12.8. Arrangement of Traffic

No traffic will be allowed on priming surface. Diversion road if required will be provided as required under Clause 112, of the MORTH specifications.

2.1.13. Methodology for Bituminous Concrete (BC)

2.1.13.1. Scope

This work shall consist of laying, in a single course, a 30 mm to 50 mm thick layer of BC on a previously prepared bituminous bound surface, as per specified lines & grades and in accordance with clause 507 of MORTH specifications.

2.1.13.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.1.13.3. Equipment's

1. Crusher of required capacity for Aggregates
2. Hot Mix Plant
3. Paver with sensor arrangement
4. Steel Tandem Rollers with required weight
5. Pneumatic Tandem Roller with required weight
6. Bitumen Sprayer
7. Dumpers
8. Air compressor
9. Pegs for putting levels
10. Broom (Mechanical)
11. Straight edge
12. Thermometers (0 - 250ºC range)
13. Gauge for checking thickness
14. Survey Kit / instrument
15. Measuring tape (metal)
16. Core cutting machine

2.1.13.4. Materials

The source and test report of all material shall be submitted to the AE for approval prior to use in aforesaid work.

1. Bitumen: The type & grade of bitumen shall be as specified in the CA or PMB conforming to IS 15462 or VG 40 / VG 30 conforming to IS 73.

2. Coarse Aggregates: The coarse aggregates shall consist of crushed rock, crushed gravel or other hard material retained on 2.36 mm sieve. It shall be clean, hard, durable and cubical shape, free from dust and soft organic and other deleterious substances. The aggregate shall satisfy the physical requirements specified in Table 500-16 of MORTH specifications and reproduced below. Where crushed gravel is proposed for use as aggregate, not less than 90% by weight of the crushed material retained on 4.75 mm sieve shall have at least two fractured faces resulting from crushing operation. Coarse aggregates shall be procured from the approved quarry and crusher. Before approval of the source, the aggregates shall be tested for stripping. Where the selected source of aggregates have poor affinity for bitumen, the bitumen shall be treated with approved anti-stripping agents, as per the manufacturer's recommendations.

Property	Test	Specification	Testing Method
Cleanliness	Grain Size analysis	Max 5% passing 0.075mm Sieve	IS: 2386 Part-I
Particle Shape	Flakiness and Elongation Index(Combined)	Max 35%	IS: 2386 Part-I
Strength	Los Angeles Abrasion Value or Aggregate Impact Value	Max 30% Max 24%	IS: 2386 Part-IV IS: 386 Part-IV
Durability	Soundness either Sodium Sulphate or Magnesium Sulphate	Max 12% Max 18%	IS: 2386 Part-V IS: 2386 Part-V
Water Absorption	Water Absorption	Max 2%	IS: 2386 Part III
Stripping	Coating and Stripping of Bitumen Aggregate	Min. Retained Coating 95%	IS: 6241
Water Sensitivity	Retained Tensile Strength	Min. 80%	AASTHO T 283

3. Fine Aggregates: Fine aggregates shall be either crushed or naturally occurring mineral material, or a combination of the two, passing the 2.36 mm sieve and retained on the 75 micron sieve. These shall be clean, hard, durable, dry and free from dust, and soft or friable matter, organic or other deleterious matter. The fine aggregate shall have a sand equivalent value of not less than 50 when tested in accordance with the requirement of IS: 2720 (part 37). The plasticity index of the fraction passing the 0.425 mm sieve shall not exceed 4, when tested in accordance with IS: 2720 (Part 5).

4. Filler: This shall conform to Clause 505.2.4 of MORTH specifications. Filler shall consist of finely divided mineral matter such as rock dust, hydrated lime or cement approved by the AE. The filler shall be graded within the limits indicated in Table 500-9 of MORTH specifications.

IS sieve (mm)	Cumulative Percent Passing by weight of Total Aggregate
0.6	100
0.3	95-100
0.075	85-100

5. Aggregate Grading and Binder Content: When tested in accordance with IS: 2386 Part I (wet sieve method), the combined grading of the coarse and fine aggregate and added filler for the particular mixture shall fall within the limits shown in Table 500-17 of MORTH specifications. Bitumen content shall be determined in accordance with clause 505.3 of MORTH specifications.

2.1.13.5. Mix Design

1. Requirement of Mix: Apart from conformity with the grading and quality requirements for individual ingredients, the mixture shall meet the requirements set out in Table 500-17 of MORTH specifications. The mix design for BC mixes using modified bitumen shall be done in the same manner as used for conventional bitumen by Marshall Method of Mix Design (MS2).
2. Mix Design: The binder content shall be optimized to achieve the requirements of the mixture set out in Table 500-17. The Marshall method for determining the optimum binder content shall be adopted as described in the Asphalt Institute Manual MS-2. The Fines to Bitumen (F/B) ratio by weight of total mix shall range from 0.6 to 1.2.
3. Job Mix Formula: The procedure for formulating the JMF shall be generally as specified in Clause 507.3.3. Mix design will be established jointly and all testing will be conducted jointly. The result thus obtained will be submitted to the AE for the approval and trial stretch will be laid based on the submitted JMF.
4. Plant Trials: After completion of JMF at lab, plant trial will be carried out to establish that the plant can be set up to produce a uniform mix conforming to the approved JMF. The permissible variation in JMF will be as per the limits specified in MORTH specifications Table 500-18.

Description	Base/Binder Course
Aggregate Passing 19mm Sieve or larger	±7%
Aggregate passing 13.2mm, 9.5mm sieve	±6%
Aggregate passing 4.75mm	±5%
Aggregate passing 2.36mm, 1.18mm, 0.6mm	±4%
Aggregate passing 0.3mm, 0.15mm	±3%
Aggregate passing 0.075mm	±1.5%
Binder content	±0.3%
Mixing Temperature	±10 ºC

2.1.13.6. Construction Operation

1. Weather and Seasonal Limitations: The work of laying shall not be taken up during rainy or foggy weather/ the base course is damp or wet or during dust storm or when the atmospheric temperature in shade is 10ºC or less as specified in the MORTH specifications.
2. Preparation of Base: The base surface on which the BC is to be laid shall be clean, and treated with Tack coat. The surface shall be thoroughly swept clean by broom and dust removed by compressed air.
3. Tack Coat: This shall be as per clause 503 of MORTH specifications.
4. Preparation and Transportation of Mix: BC shall be prepared in batch in Hot Mix Plant. The mixing will be done in a twin shaft pug mill, which will produce a homogeneous

mix. The mixture shall be transported from the batching plant in tippers / dumpers covered with tarpaulin, as and when required, so as to maintain the temperature, over spillage and dust and environment hazard.

5. Spreading: The mix shall be transported to the site by the dumpers, which shall slowly discharge the mix in the paver hopper, while both the tipper and paver will move forward steadily and slowly. The automatic sensor shall ensure that the mix is being laid to the proper line, grade and level. The direction of the paver will be guided by the string lines fixed with the stack & arms on both sides of the edges 0.5 m away from the edge. Transverse & longitudinal joints shall be cut vertically with asphalt cutter and a coat of hot bitumen shall be applied before placing materials.
6. Rolling & Compaction: As soon as the mix has been laid by the paver, compaction shall be done with the help of a set of rollers moving at a speed not exceeding 5 km/hour following close to the paver. Rolling sequence will be as established during laying trial stretch.

   Compaction shall subsequently be completed before the temperature falls below the minimum rolling temperatures. Rolling of the longitudinal joints shall be done immediately behind the paving operation. After this, rolling shall commence at the edges and progress towards the centre longitudinally except that on super elevation and unidirectional cambered portions, it shall progress from the lower to the upper edge parallel to the centre line of the pavement. Rolling shall continue until all roller marks have been removed from the surface.

   a. Initial or breakdown rolling shall be done with 80-100 kN dead weight smooth wheeled rollers.
   b. Intermediate rolling shall be done by 8-10 tons dead weight or vibratory roller or with a pneumatic tired roller of 12 to 15 tons weight having nine wheels, with a tire pressure of at least 5.6 kg/sq. cm. The finish rolling shall be done with 6 to 8 tons smooth wheeled tandem rollers and the no. of passes required shall be decided after laying a trial bed.

   During rolling, wheels of rollers shall be kept moist by light sprinkling water from the water storage fitted with the roller to prevent the mix from adhering to the wheels. The roller shall proceed on the fresh material with rear or fixed wheel leading so as to minimize the pushing of the mix.

   In portions in camber, rolling should begin at the edge with the roller forward and backward until the edges have been firmly compacted. The roller shall then progress gradually towards the centre parallel to the centre line uniformly overlapping each of the preceding tracks by one-third width until the entire surface has been rolled.

   Along forms, kerb and walls where rolling cannot be done by rollers, shall be compacted with mechanical tampers or a plate compactor or rammers as may be feasible.

   Any undulations / displacement found as a result of reversing of the direction of a roller or due to some other reasons shall be rectified as specified or removed and made good to the satisfaction of the AE.

   All the field tests such as checking of compaction by core cutting, surface irregularity by using a straight edge and checking of cross slope by taking level, rate of spray of

tack coat using trays, temperature of mix using digital thermometer, depth of the layer using thickness gauge etc. shall be carried out as per the requirement of MORTH specifications clause 900.

The compacted layers of BC shall have a minimum field density equal to or more than 92% of the average theoretical maximum specific gravity (G_{mm}) obtained on the day of compaction.

2.1.13.7. Opening to Traffic

As per MORTH specifications Clause 507.5.

2.1.13.8. Environmental Control

The work shall be executed with the least disturbance on environment.

1. The excess material and any other excess materials shall be disposed suitably without polluting streams, water courses.
2. Vehicles tracks shall be kept moist to prevent flying of dust particles with the vehicle movement.
3. The smoke and dust at plant will be controlled by adopting suitable method.

2.1.13.9. Surface Finish and Quality of work

The surface finish of the completed construction shall conform to the requirement of clause 902. For control of the quality of materials and the work carried out, the provisions of section 900 shall apply.

2.1.14. Methodology for Dry Lean Concrete (DLC)

2.1.14.1. Scope of work

The work shall consist of construction of (zero slump) DLC sub-base for cement concrete pavement in accordance with the requirements of MORTH specifications clause 601 and in conformity with the lines, grades and cross sections shown on the drawings.

2.1.14.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. IRC SP: 49-2014
4. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
5. Relevant Drawings

2.1.14.3. Material

The Contractor shall indicate to the AE the source of all materials with relevant test data to be used in DLC work sufficiently in advance and the approval of the AE for the same shall be obtained before the schedule commencement of the work in trial length.

1. Cement: Cement to be used may preferably be obtained in bulk form. Cement shall be transported, handled and stored on the site in such a manner as to avoid deterioration or contamination. Cement shall be stored above ground level in perfectly dry and water tight sheds and shall be stacked to a height of not more than eight bags. Wherever bulk storage containers are used, their capacity should be sufficient to cater the requirement at site. The containers shall be cleaned at least once every 3 months. MTC or lab test certificate shall be maintained for each and every batch of cement received.

Following type of cement may be used with prior approval of the AE.

S. No	Type of Cement	Conforming to
1	OPC 43 Grade	IS:269
2	Portland Slag Cement	IS: 455
3	Portland Pozzolana Cement	IS: 1498-Part I

If subgrade soil contains soluble sulphates in a concentration more than 0.5%, sulphate resistant cement conforming to IS: 12330 or Portland slag cement with slag up to 50% shall be used.

2. Fly-Ash: Fly-Ash up to 30 percent by weight of cementitious material (Cement + Fly-ash) may be used cement may be used to replace OPC cement and in such case the cement content shall not be less than 120 kg/cum. Fly-ash shall conform to IS: 3812 Part I.

3. Aggregates: Aggregates for DLC shall be natural material complying with IS: 383. The aggregates shall not be alkali reactive. The limits of deleterious materials shall not exceed the requirements set forth in table 600-2 of MORTH specifications.

   a. Coarse Aggregates: Coarse Aggregate shall comply with clause 602.2.6.2, except that the maximum size of the coarse aggregate shall be 26.5 mm, and aggregate gradation shall comply with Table 600-1.
   b. Fine Aggregates: The fine aggregate shall comply with clause 602.2.6.3.

4. Curing Compound: The Curing compound shall be white pigmented or transparent type with water retention index of 90 % when tested in accordance with BS 7542.
5. Water: Water used for mixing and curing shall comply with Clause 602.2.7.

2.1.14.4. Proportioning of Materials for the Mix

The mix shall be proportioned with a maximum aggregate cementitious material ratio of 14:1 (as per IRC SP 49) where OPC is used and 12:1 where PPC or PSC is used. The water content shall be adjusted to the optimum as per clause 601.3.2 for facilitating compaction by rolling. The strength and density requirements of concrete shall be determined in accordance with clause 601.7 and clause 601.8 by making trial mixes. Care should be taken to prevent one size of aggregate falling into other size of the hopper of the feeding bin while loading the individual size of the aggregates into the bins.

1. Moisture Content: The OMC shall be determined and demonstrated by rolling during trial length construction and the OMC and degree of compaction shall be got approved from AE. While laying in the main work, the DLC shall have a moisture content between the optimum and optimum +2 percent, keeping in view the effectiveness of the compaction achieved and to compensate for evaporation losses.

2. Cement Content: The cement content in the DLC shall be such that the strength specified in clause 4.1 of IRC SP: 49 is achieved. The minimum cementitious content shall not be less than 140 kg/cum of concrete. In case fly-ash is blended at site as part replacement of cement, the quantity of fly-ash shall be 15-30 percent by weight of cementitious material and the content of OPC shall not be less than 100 kg/cu. m.

3. Concrete Strength: The average compressive strength of each consecutive group of 5 concrete cubes shall not be less than 10 MPa at 7 days. In addition, the compressive strength of any individual concrete cube shall not be less than 7.5 MPa at 7 days.

4. Sub-Grade: The subgrade shall conform to the grades and cross-sections on the drawings and shall be uniformly compacted to the modified Proctor density not less than 97%. The lean concrete sub-base shall not be laid on a subgrade softened by rain after its final preparation, surface trenches and soft spots, if any, must be properly back-filled and compacted to avoid any weak spot. A day before placing of the sub-base, the subgrade surface shall be given a fine spray of water and rolled with one or two passes of a smooth wheeled roller after a lapse of 2-3 hours in order to stabilise loose surface. If found necessary, another fine spray of water may be applied just before placing the sub-base.

5. Drainage Layer: A drainage layer conforming to Clause 401 shall be laid above the subgrade before laying the DLC sub-base.

2.1.14.5. Construction

1. General: The DLC shall be laid on the prepared granular drainage layer. The DLC sub-base shall be overlaid with concrete pavement only after 7 days of sub-base construction.

2. Batching and Mixing: The batching plant shall be capable of proportioning the materials by weight, each type of material being weighed separately in accordance with Clause 602.9.2, 602.9.3.1 and 602.9.3.2 of MORTH specifications.

3. Transporting: Plant mix DLC shall be discharged immediately from the mixer, transported directly to the point where it is to be laid and protected from the weather by covering the tipping trucks with tarpaulin during transit if necessary.

4. Placing: DLC shall be placed by a paver with electronic sensor on the drainage layer. In roads with carriageway more than 2 Lanes a longitudinal joint shall be provided. Transverse Butt type joints shall be provided at the end of the construction in a day. The DLC shall be laid in such a way that it is at least 1000 mm wider on each side than the proposed width including paved shoulders of the concrete pavement.

5. Compaction: The compaction shall be carried out immediately after the material is laid and levelled. In order to ensure thorough compaction, rolling shall be continued on the full width till there is no further visible movement under the roller and surface is well closed. The minimum dry density obtained shall not be less than 98 percent of that achieved during the trial length construction in accordance with Clause 601.7. The densities achieved at the edges i.e. 0.5 m from the edge shall not be less than 96 percent of that achieved during the trial construction. Double drum smooth-wheeled vibratory rollers of minimum 80 to 100 kN static weight are suitable for rolling DLC.

   The spreading, compacting and finishing of the lean concrete shall be carried out as rapidly as possible and the operation shall be so arranged as to ensure that the time between the mixing of the first batch of concrete in any transverse section of the layer and the final finishing of the same shall not exceed 90 minutes when the temperature of concrete is between 25°C and 30°C, and 120 minutes if less than 25°C. This period

may be reviewed by the Engineer in the light of the results of the trial run but in no case shall it exceed 120 minutes. Work shall not proceed when the temperature of the concrete exceeds 30°C. If necessary, chilled water or addition of ice may be resorted to for bringing down the temperature. It is desirable to stop concreting when the ambient temperature is above 35°C. After compaction has been completed, roller shall not stand on the compacted surface for the duration of the curing period except during commencement of next day's work near the location where work was terminated the previous day.

6. Joints: Construction and Longitudinal joints shall be provided as per drawings. Transverse butt type joint shall be provided at the end of the construction in a day. Longitudinal construction joint shall be provided only when full width paving is not possible. Transverse joints in DLC shall be staggered from the construction butt type joint in concrete pavement by 800-1000 mm. Longitudinal joint in DLC shall be staggered by 300-400 mm from the longitudinal joint of concrete pavement.

7. Curing: The curing shall be done by either of the following ways:
    a. Curing may be done by covering the surface by gunny bags / hessian, which shall be kept wet continuously for 7 days by sprinkling water.
    b. The curing shall be done by spraying with approved resin based aluminized reflective curing compound in accordance with clause 602.9.12. As soon as the curing compound has lost its tackiness, the surface shall be covered with wet hessian for 3 days.
    c. Wax-based white pigmented curing compound with water retention index of less than 90% shall be used to cure the dry lean concrete. The compound shall be applied uniformly with a mechanical sprayer and with a hood to protect spray from the wind.

2.1.14.6. Quality Control

Control of quality of materials and works shall be exercised by the Engineer in accordance with Section 900.

2.1.15. Methodology for Pavement Quality Concrete (PQC)

2.1.15.1. Scope

The work shall consist of construction of un-reinforced, dowel jointed, plain cement concrete pavement in accordance with the requirements of MORTH specifications and inconformity with the lines, grades and cross sections shown on the drawings.

2.1.15.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. IRC: 58-2015
4. IRC: 15-2017
5. IRC: 44-2017
6. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
7. Relevant Drawings

2.1.15.3. Materials

1. Source of Materials: The Contractor shall indicate to the AE the source of all materials to be used in the concrete work with relevant test data sufficiently in advance.

2. Cement: Any of the following types of cement capable of achieving the design strength may be used.

S. No	Type of Cement	Conforming to
1	OPC 43 Grade	IS:269
2	OPC 53 Grade	IS:269
3	Portland Slag Cement	IS: 455
4	Portland Pozzolana Cement	IS: 1498-Part I

If the soil around concrete pavement has soluble salts like sulphates in excess of 0.5 percent, the cement used shall be sulphate resistant and shall conform to IS: 12330.

3. Admixtures:

   a. Chemical Admixtures: Admixtures conforming to IS: 9103 and IS: 6925 shall be permitted to improve workability of concrete or extension of setting time, on satisfactory evidence that they will not have any adverse effect on the properties of concrete with respect to strength, volume change, durability and have no deleterious effect on steel bars.

   b. Mineral Admixtures: Fly-ash up to 25% by weight of cementitious material may be used in Ordinary Portland cement 43 or 53 grade as part replacement of cement provided uniform blending with cement is ensured. The fly ash shall conform to IS: 3812 (Part I). Site mixing of fly ash shall be permitted, ensuring availability of the equipment's at site for uniform blending through a specific mechanised facility with automated process control like batch mix plants conforming to IS:4925 and IS:4926.

4. Fibres: Fibres may be used subject to the provision in the design / approval by the AE to reduce the shrinkage cracking and improve the post-cracking residual strength. The fibres may be steel fibre as per IRC SP: 46 or polymeric Synthetic fibres within the following range of specifications:

Effective Diameter	10 micron-100 micron
Length	6-48 mm
Specific Gravity	More than 1.0
Suggested Dosage	0.6-2.0 kg/cu. m (0.2-0.6% by weight of cement in mix) Usage will be regulated as stipulated in IRC:44/IS:456
Water Absorption	Less than 0.45 percent.
Melting Point of this fibre shall not be less than 160ºC	
The aspect ratio generally varies from 200 to 2000	
Synthetic fibres will have good alkali and UV light resistance.	
Slump of concrete at paving site shall be 25±15 mm for pave construction or 40±15 mm for manual construction	

5. Aggregate: Aggregates for pavement concrete shall be natural material complying with IS: 383 but with Los Angeles Abrasion Test value not exceeding 35 percent. The limits of deleterious materials shall not exceed the requirements set out in Table 600-2.

    a. Coarse Aggregate: The maximum size of coarse aggregate shall not exceed 31.5 mm for pavement concrete. The Los Angeles Abrasion Value shall not exceed 35. The combined flakiness and elongation index of aggregate shall not be more than 35 percent.
    b. Fine Aggregate: The Fine aggregate shall consist of clean natural sand or crushed stone sand or a combination of the two and shall conform to IS: 383. The fine aggregates shall have a sand equivalent value of not less than 50 when tested in accordance with requirement of IS: 2720 (Part-37)
    c. Combined Gradation of Fine and Coarse Aggregate: The combined gradation of fine and coarse aggregates shall be as per Table 600-3 of MORTH specifications.

6. Water: Water used for mixing and curing of concrete shall be clean and free from injurious amount of oil, salt, acid, vegetable matter and other substances harmful to the finished concrete.

7. Steel for Dowels and Tie Bars: Steel shall conform to the requirements of IS: 432 and IS: 1786 as relevant. The dowel bars shall conform to IS: 432 of Grade I. Tie bars shall be either High Yield Strength Deformed bars conforming to IS: 1786 and grade of Fe 500 or plain bars conforming to IS: 432 of Grade I. The steel shall be coated with epoxy paint for protection against corrosion.

8. Joint Filler Board: Synthetic Joint filler board for expansion joints shall be used only at abutting structures like bridges and shall be of 20-25 mm thickness with a tolerance of ±1.5 mm and of a firm compressible material and complying with the requirements of IS: 1838, with a compressibility more than 25%.

9. Joint Sealing Compound: The joint sealing compound shall be of hot poured, elastomeric type or cold polysulphide / polyurethane / silicone type having flexibility, resistance to age hardening and durability as per IRC: 57.

10. Preformed Seals: The pre-formed sealing material shall be vulcanized elastomeric compound using polychloroprene (Neoprene) as the base polymer.

11. Storage of Materials: All materials shall be stored in accordance with the provisions of Clause 1014 of MORTH specifications.

2.1.15.4. Proportioning of Concrete

1. Cement Content: The quantity of cement shall not be less than 360 kg/cu. m for all types of cement. In case fly ash grade I (as per IS:3812) is blended at site as part replacement of cement, the quantity of fly ash shall be up to 25% by weight of cementitious material and the quantity of OPC in such a blend shall not be less than 310 kg/cu. m. The OPC content in any case shall not be more than 250 kg/cum.

2. Concrete Strength: The characteristic flexural strength of concrete for 28 days shall not be less than 4.5 MPa unless specified otherwise. Target mean flexural strength for mix

design shall be more than 4.5 MPa + 1.65s, where "s" is standard deviation of flexural strength derived by conducting test on minimum 30 beams.

The water content shall be the minimum required to provide the agreed workability for full compaction of the concrete to the required density as determined by the trial mixes and the maximum free water cement ratio shall be 0.40 when only OPC is used and 0.50 when blended cement (Portland Pozzolana Cement or Portland Slag Cement or OPC blended with fly ash or Ground Granulated Blast Furnace Slag, at site) is used IRC 44 shall also be referred regarding the selection of w/c ratio.

The ratio between the 7 and 28 day strength shall be established for the mix to be used in the slab in advance, by testing pairs of beams and cubes at each stage on at least six batches of trial mix. The average strength of the 7 day cured specimens shall be divided by the average strength of the 28 day specimens for each batch, and the ratio "R" shall be determined. The ratio "R" shall be expressed to three decimal places.

3. Workability: The workability of the concrete at the point of placing shall be adequate for the concrete to be fully compacted and finished without undue flow. The optimum workability for the mix to suit the paving plant being used shall be determined by the Contractor and approved by the AE. The workability shall be established for the type of paving equipment available. A slump value in the range of 25 ± 15 mm is reasonable for paving works but this may be modified depending upon the site requirement and got reviewed by the AE.

4. Design Mix: The laboratory trials of design mix shall be carried out with the materials from the approved sources to be used as per IRC: 44. Trial mixes shall be made in presence of the AE and the design mix shall be subject to the review of AE.

5. Sub-Base: The cement concrete pavement shall be laid over the DLC sub-base constructed in accordance with the relevant drawings and specifications. It shall be ensured that the DLC sub-base is not damaged before laying the concrete pavement. If the DLC is found damaged at some places or it has cracks wider than 10 mm, it shall be repaired with fine cement concrete (aggregate size 10 mm and down) or bituminous concrete before laying separation membrane layer.

6. Separation Membrane: A separation membrane shall be used between the PQC and the DLC sub-base. Separation membrane shall be impermeable PVC sheet of 125 micron thickness, transparent or white in colour laid flat with minimum creases. Before placing the separation membrane, the DLC sub-base shall be swept clean of all the extraneous materials using air compressor. Wherever overlap of plastic sheets is necessary, the same shall be at least 300 mm and any damaged sheathing shall be replaced. The separation membrane may be nailed to the lower layer with concrete nails. The separation membrane shall be omitted when two layers of wax-based curing compound is used.

7. Tie Bars: Tie bars shall be provided at the longitudinal joints as per dimensions and spacing shown in the drawing and in accordance with Clause 602.6.6. The direction of the tie bars at curves shall be radial in the direction of the radius.

8. Dowel Bars: The work shall be done in accordance to the clause 602.6.5 of MORTH specifications. Dowel Bars shall be mild steel rounds in accordance with Clause

602.2.8, details / dimensions as indicated in the drawings and free from oil, dirt, loose rust or scale. They shall be straight, free of irregularities and burring restricting slippage in the concrete. The sliding ends shall be sawn or cropped cleanly with no protrusions outside the normal diameter of the bar.

Unless shown otherwise on the drawings, dowel bars shall be positioned at mid depth of the slab within a tolerance of ±20 mm, and centred equally about intended lines of the joint within a tolerance of ±25 mm.

a. For bars supported on cradles prior to the laying of the slab.
    i. All bars in a joint shall be within ±2 mm per 300 mm length of bar.
    ii. 2/3rd of the number of bars shall be within ±3 mm per 500 mm Length of bar
    iii. No bar shall differ in alignment from an adjoining bar by more than 3 mm per 300 mm length of bar in either the horizontal or vertical plane.
    iv. Cradles supporting dowel bar shall not extend across the line of joint i.e. no steel bar of the cradle assembly shall be continuous across the joint.

b. For all bars inserted after laying of the slab except those inserted by a Dowel Bar Inserter the tolerance for alignment may be twice as indicated in (a) above.

The assembly of dowel bars and supporting cradles, including the joint filler board in the case of expansion joints, shall have the following degree of rigidity when fixed in position:

a. For expansion joints, the deflection of the top edge of the filler board shall be not greater than 13 mm, when a load of 1.3 kN is applied perpendicular to the vertical face of the joint filler board and distributed over a length of 600 mm by means of a bar or timber packing, at mid depth and midway between individual fixings, or 300 mm from either end of any length of filler board, if a continuous fixing is used. The residual deflection after load shall be not more than 3 mm.

b. The fixings for joint assembly shall not fail under 1.3 kN load and shall fail before the load reaches 2.6 kN when applied over a length of 600 mm by means of a bar or timber packing placed as near to the level of the line of fixings as practicable.

c. Fixings shall be deemed to fail when there is displacement of the assemblies by more than 3 mm with any form of fixing, under the test load. The displacement shall be measured at the nearest part of the assembly to the centre of the bar or timber packing.

Dowel bars in the contraction joints, construction joints and expansion joints shall be covered by a thin plastic sheath. The thickness of the sheath shall not exceed 0.5 mm and shall be tightly fitted on the bar for at least two-thirds of the length from one end for dowel bars in contraction/construction joints and half the length plus 50 mm for expansion joints.

2.1.15.5. Joints

The locations and type of joints shall be as shown in the drawing. Joints shall be constructed depending upon their functional requirement. The location of the joints should be transferred accurately at the site and mechanical saw cutting of joints done as per stipulated dimensions. It shall be ensured that the required depth of cut is made from edge-to-edge of the pavement. Transverse and longitudinal joints in the PQC and DLC shall be staggered so that they are not coincident vertically and are at least 800 to 1000 mm and 300 to 400 mm apart respectively. Sawing of joints shall be carried out with diamond studded blades soon after the concrete has hardened to take the load of the sewing machine and crew members without damaging the texture of the pavement.

Sawing operation could start as early as 4-8 hours after laying of concrete pavement but not later than 8 to 12 hours depending upon the ambient temperature, wind velocity, relative humidity and required maturity of concrete achieved for this purpose.

When the Kerb is cast integrally with the main pavement slab, the joint cutting shall also be extended to the kerb.

1. Transverse Joints: Transverse joints shall be contraction, construction and expansion joints constructed at the spacing described in the drawings. Transverse joints shall be straight within the following tolerances along the intended line of joints.

    a. Deviations of the performed filler board (IS: 1838) in the case of expansion joints from the intended line of the joint shall not be greater than ±10 mm.

    b. The best fit straight line through the joint grooves as constructed shall be not more than 25 mm from the intended line of the joint.

    c. Deviations of the joint groove from the best fit straight line of the joint shall not be greater than 10 mm.

    d. Transverse joints on each side of the longitudinal joint shall be in line with each other and of the same type and width. Transverse joints shall have a sealing groove which shall be sealed in compliance with Clause 602.10.

2. Contraction Joints: The contraction joints shall be placed transversely at pre-specified locations as per drawings / design using dowel bars. These joints shall be cut as soon as the concrete has undergone initial hardening and is hard enough to take the load of joint sawing machine without causing damage to the slab. Contraction joints shall consist of a mechanical sawn joint groove, 3 to 5 mm wide and one-fourth to one-third depth of the slab ± 5 mm or as stipulated in the drawings and dowel bars complying with Clause 602.6.5.

3. Expansion Joints: The expansion joint shall consist of a joint filler board complying with Clause 602.2.9 and dowel bars complying with Clause 602.6.5 and as detailed in the drawings. The filler board shall be positioned vertically with the prefabricated joint assemblies along the line of the joint within the tolerances given in Clause 602. 6.2.1. The adjacent slabs shall be completely separated from each other by the joint filler board.

4. Transverse Construction Joints: Transverse construction joint shall be placed whenever concreting is completed after a day's work or is suspended for more than 30 minutes. These joints shall be provided at location of contraction joints using dowel bars. If sufficient concrete has not been mixed to form a slab extending up to a contraction joint, and if an interruption occurs, the concrete placed shall be removed up to the last preceding joint and disposed of.

5. Longitudinal Joint: The longitudinal joints shall be constructed by forming or by sawing as per details of the joints shown in the drawing. Sawed longitudinal joints shall be constructed when the concrete pavement placement width exceeds 4.5 m. The groove may be cut after the final set of the concrete. Joints should be sawn to at least one-third the depth of the slab ±5 mm as indicated in the drawing.

2.1.15.6. Concreting during Monsoons

Concreting should be avoided during rainy season. However, when concrete is being placed during monsoon months and when it may be expected to rain, sufficient supply of tarpaulin or other waterproof cloth shall be provided along the line of the work. Any time when it rains, all freshly laid concrete which had not been covered for curing purposes shall be adequately protected. Any concrete damaged by rain shall be removed and replaced.

2.1.15.7. Temperature Limitation

No concreting shall be done when the temperature of the concrete reaching the paving site is above 30°C. Besides, in adverse conditions like high temperature, low relative humidity, excessive wind velocity, imminence of rains etc., tents on mobile trusses may be provided over the freshly laid concrete for a minimum period of 3 hours as directed by the AE. To bring down the temperature, if necessary, chilled water or ice flakes should be made use of. When the ambient temperature is more than 35°C, no concreting shall be permitted.

2.1.15.8. Fixed Form Paving

1. Side Forms and Rails: These shall be provided in case of fixed form paving. All side forms shall be of mild steel of depth equal to the thickness of pavement or slightly less to accommodate the surface irregularity of the sub-base. The forms can be placed in series of steel packing plates or shims to take care of irregularity of sub-base.

   The forms and rails shall be firmly secured in position by not less than 3 stakes/pins for every 3 m length so as to prevent movement in any direction. Forms and rails shall be straight within a tolerance of 3 mm in 3 m and when in place shall not settle in excess of 1.5 mm in 3 m while paving is being done.

   The forms shall be bedded on a continuous bed of low moisture content lean cement mortar or concrete and set to the line and levels shown on the drawings within tolerances ±10 mm and ±3 mm respectively. At all times sufficient forms shall be used and set to the required alignment for at least 300 m. Length of pavement immediately in advance of the paving operations, or the anticipated length of pavement to be laid within the next 24 hours whichever is more.

2. Slip Form Paving:
   a. Use of Guidewires: Where slip form paving is proposed, a guidewire shall be provided along both sides of the slab. Each guidewire shall be at a constant height above and parallel to the required edges of the slab as described in the contract drawing within a vertical tolerance of ±3 mm. Additionally, one of the

wires shall be kept at a constant horizontal distance from the required edge of the pavement as indicated in the contract drawing within a lateral tolerance of ±10 mm. The guidewires shall be supported on stakes 5-6 m apart by connectors capable of fine horizontal and vertical adjustment. The guidewire shall be tensioned on the stakes so that a 500 gm weight shall produce a deflection of not more than 20 mm when suspended at the midpoint between any pair of stakes. The ends of the guidewires shall be anchored to fixing point or winch and not on the stakes. On the curves, the stakes shall be fixed at not more than 3 m Centre-to-Centre. The stakes shall be positioned and hammered into the ground and the connectors will be maintained at their correct height and alignment from 12 hours on the day before concreting takes place till after finishing of texturing and spraying of curing compound on the concrete.

3. Construction
   a. General: A systems approach may be adopted for construction of the pavement, and the Method Statement for carrying out the work, detailing all the activities, indication of time-cycle, equipment, personnel etc., shall be got approved from the AE before the commencement of the work. This shall include the type, capacity and make of the batching and mixing plant besides the hauling arrangement and paving equipment. The capacity of paving equipment, batching plant as well as all the ancillary equipment shall be adequate for a paving rate of at least 500 m in one day. The paving speed of slip-form paver shall not be less than 1.0 m per minute. The concreting should proceed continuously without stops and starts.

   b. Batching and Mixing: Batching and mixing of the concrete shall be done at a central batching and mixing plant, with automatic controls, located at a suitable place which takes into account sufficient space for stockpiling of cement, aggregates and stationary water tanks. This shall be located at an approved distance, duly considering the properties of the mix and the transporting arrangements available with the Contractor.

   c. Equipment for Proportioning of Materials and Paving: Proportioning of materials shall be done in the batching plant by weight, each type of material being weighed separately. The cement from the bulk stock may be weighed separately from the aggregates. Water shall be measured by volume. Specified percentage of plasticizer in volume will be added by weight of cement. Wherever properly graded aggregate of uniform quality cannot be maintained as envisaged in the mix design, the grading of aggregates shall be controlled by appropriate blending techniques.

      i. Batching Plant and Equipment:
         - General: The batching plant shall include minimum four bins, weighing hoppers, and scales for the fine aggregates and for each size of coarse aggregate. If cement is used in bulk, a separate scale for cement shall be included. There shall be a separate bin for fly ash, if this additive is specified. The weighing hoppers shall be properly sealed and vented to preclude dust during operation. The batch plant shall be equipped with a suitable non-resettable batch counter which will correctly indicate the number of batches proportioned. A continuous type of mixing plant can also be used

provided the ingredients are weighed through electronic sensors before feeding.

- Automatic weighing devices: Batching plant shall be equipped to proportion aggregates and bulk cement by means of automatic weighing devices using load cells.
- Mixer: Mixers shall be pan type, reversible type or any other mixer capable of combining the aggregates, cement, and water into a thoroughly mixed and uniform mass within the specified mixing period, and of discharging the mix, without segregation. Each stationary mixer shall be equipped with an approved timing device which will automatically lock the discharge lever when the drum has been charged and release it at the end of the mixing period. The pick-up and throw• over blades in the drum or drums shall be repaired or replaced when they are worn down 20 mm or more. The Contractor shall (1) have available at the job site a copy of the manufacturer's design, showing dimensions and arrangements of blades in reference to original height and depth, or (2) provide permanent marks on blade to show points of 20 mm wear from new conditions.
- Control cabin: An air-conditioned centralized computer control cabin shall be provided for automatic operation of the equipment.
- The design features of the batching plant should be such that it can be shifted quickly.

ii. Paving Equipment: The concrete shall be placed with an approved fixed form or slip form paver with independent units designed to (i) spread, (ii) consolidate, screed and float-finish, (iii) texture and cure the freshly placed concrete in one complete pass of the machine in such a manner that a minimum of hand finishing will be necessary and so as to provide a dense and homogeneous pavement in conformity with the plans and Specifications.

Vibrators shall operate at a frequency of 8000-10000 impulses per minute under load at a maximum spacing of 600 mm. The variable vibration setting shall be provided in the machine.

iii. Concrete Stew: The Contractor shall provide adequate number of concrete saws with sufficient number of diamond-edge saw blades. The saw machine shall be either electric or petrol / diesel driven type. A water tank with flexible hose and pump shall be made available for this activity on priority basis.

d. Hauling and Placing of Concrete: Freshly mixed concrete from the central batching and mixing plant shall be transported to the paver site by means of tipping trucks or transit mixers of sufficient capacity and approved design in sufficient numbers to ensure a constant supply of concrete. Covers shall be used for protection of concrete against the weather. While loading the concrete truck shall be moved back and forth under the discharge chute to prevent segregation. The tipping trucks shall be capable of maintaining the mixed concrete in a homogeneous state and discharging the same without segregation and loss of cement slurry.

i. Placing of Concrete: The total time taken from the addition of the water to the mix, until the completion of the surface finishing and texturing shall not exceed 120 minutes when concrete temperature is less than 25°C and 90 minutes when the concrete temperature is between 25°C and 30°C.

The placing of concrete in front of the PQC paver should preferably be from the side placer to avoid damage to DLC by concrete tipping trucks. In case of unavoidable situation, truck supplying concrete to the paver may be allowed to ply on the DLC with the approval of the Engineer.

The paver shall be capable of paving the carriageway as shown in the drawings, in a single pass and lift.

Where fixed form pavers are to be used, forms shall be fixed in advance as per Clause 602.8. Before any paving is done, the site shall be shown to the AE, in order to verify the arrangement for paving besides placing of dowels, tie-bars etc., as per the relevant Clauses of these Specifications. The mixing and placing of concrete shall progress only at such a rate as to permit proper finishing, protecting and curing of the concrete in the pavement.

While the concrete is still plastic, its surface shall be textured by brush or tines as per the instructions of the AE in compliance with Clause 602.9.11. The surface and edges of the slab shall be cured by the application of a sprayed liquid curing membrane in compliance with Clause 602.9.12. After the surface texturing, but before the curing compound is applied, the concrete slab shall be marked with the chainage at every 100 m interval by embossing.

e. Construction by Slip Form Paver: The slip form paving train shall consist of a power machine which spreads, compacts and finishes the concrete in a continuous operation. The slip form paving machine shall compact the concrete by internal vibration and shape it between the sides forms with either a conforming plate or by vibrating and oscillating finishing beams. The concrete shall be deposited without segregation in front of slip form paver across the whole width and to a height which at all times is in excess of the required surcharge. The deposited concrete shall be struck off to the necessary average and differential surcharge by means of the strike off plate or a screw auger device extending across the whole width of the slab. The equipment for striking-off the concrete shall be capable of being rapidly adjusted for changes of the average and differential surcharge necessitated by change in slab thickness or cross fall.

The level of the conforming plate and finishing beams shall be controlled automatically from the guide wires installed as per Clause 602.8 by sensors attached at the four corners of the slip form paving machine. The alignment of the paver shall be controlled automatically from the guide wire by at least one set of sensors attached to the paver. The alignment and level of ancillary machines for finishing, texturing and curing of the concrete shall be automatically controlled relative to the guide wire or to the surface and edge of the slab.

Slip-form paving machines shall have vibrators of variable output, with a maximum energy output of not less than 2.5 KW per metre width of slab per 300mm depth of slab for a laying speed up to 1.5 m per minute. The machines shall be of sufficient mass to provide adequate reaction during spreading and paving operations on the traction units to maintain forward movements during the placing of concrete in all situations. Normal paving speed shall be maintained as per Clause 602.9.1.

Slip-form pavers with adequate width to pave the entire carriageway width in one go shall be employed unless specified in the Contract. In situations where full-width paving is not possible, paving in part widths may be permitted by the Engineer. Paving in part will be avoided, except in unavoidable circumstances. In case of part width paving, care shall be taken to ensure that while laying the next lane, bond between the remaining half length of tie bar or subsequently inserted tie bars and the newly laid concrete is adequately developed. Care shall be taken to avoid damage to the previous lane.

i. Tube Floating: Upon the instructions of the Engineer, Contractor shall scrape the concrete surface when in plastic state with a 3 m long tube float fixed with a long and stable handle before texturing. Tube float shall be of an alloy steel tube of 50 to 60 mm diameter with a long and stable handle. The length of tube float shall preferably be longer than half the length of slab i.e. half the distance between two transverse contraction joints.

   The tube float shall be placed at the Centre of the slab to longitudinal joint and pulled slowly and uniformly towards the edges. After the use of float tube, it shall be frequently cleaned before further use. The slurry removed shall be discarded.

f. Construction by Fixed Form Paver: The fixed form paving train shall consist of separate powered machines which spread, compact and finish the concrete in a continuous operation.

   The concrete shall be discharged without segregation into a hopper spreader which is equipped with means for controlling its rate of deposition on to the sub-base. The spreader shall be operated to strike off concrete up to a level requiring a small amount of cutting down by the distributor of the spreader. The distributor of spreader shall strike off the concrete to the surcharge adequate to ensure that the vibratory compactor thoroughly compacts the layer.

g. Semi-Mechanized Construction: Areas in which hand-guided methods of construction become indispensable shall be got approved by the AE in writing in advance. Such work may be permitted only in restricted the areas in small lengths.

h. Transition Slabs: At the he interface of rigid and flexible pavement, at least 3 m long reinforced buried slab shall be provided to give a long lasting joint at the interface. The details shall be as given in IRC: 15.

i. Surface Texture:
   i. Tining: After final floating and finishing of the slab and before application of the liquid curing membrane, the surface of concrete slabs shall be textured either in the transverse direction (i.e. at right angles to the

longitudinal axis of the road) or in longitudinal direction (i.e. parallel to the centre line of the roadway). The texturing shall be done by tining the finished concrete surface by using rectangular steel tines. A beam or a bridge mounted with steel tines shall be equipped and operated with automatic sensing and control devices from main paver or auxiliary unit.

ii. Brush Texturing: Alternatively on the instructions of the AE, the brush texturing shall be applied. The brushed surface texture shall be applied evenly across the slab in one direction by the use of a wire brush not less than 450 mm wide but wider brushes normally of 3 m length are preferred. The brush shall be made of 32 gauge tape wires grouped together in tufts placed at 10mm centres. The tufts shall contain an average of 14 wires and initially be 100 mm long. The brush shall have two rows of tufts. The rows shall be 20 mm apart and the tufts in one row shall be opposite the Centre of the gap between tufts in the other row. The brush shall be replaced when the shortest tuft wears down to 90 mm long.

j. Curing: Immediately after the surface texturing, the surface and sides of the slab shall be cured by the application of approved resin-based aluminized reflective curing compound which hardens into an impervious film or membrane with the help of mechanical sprayer. The curing compound shall not react chemically with the concrete and the film or membrane shall not crack, peel or disintegrate within three weeks of application.

k. Preparation and Sealing of Joint Grooves: All Joints shall be sealed using sealants described in Clause 602.2.10 and in accordance with clause 602.10

2.1.16. Methodology for Thermoplastic Road Marking Paint

2.1.16.1. Scope

The work shall consist of providing and laying of pavement marking with hot applied thermoplastic paints (type-2) as per Clause 803 of MORTH specifications.

2.1.16.2. General

This method statement covers providing and laying of pavement marking with hot applied thermoplastic paints as specified in drawing or as per IRC: 35.

2.1.16.3. Materials

Materials of road marking shall be hot applied thermoplastic compound and the same shall meet the requirements as specified in clause 803.4 of MORTH specifications.

1. Composition: The pigment, beads, and aggregate shall be uniformly dispersed in the resin. The material shall be free from all skins, dirt and foreign objects and shall comply with requirements indicated following table.

   Proportions of constituents of Material (% by weight) (Table 800-9 of MORTH specifications)

Component	White	Yellow
Binder	18 min	18 min

Glass Beads	30-30	30-30
Titanium Dioxide	10 min	-
Calcium Carbonate and Inert Fillers	42 max	Amount of yellow pigment, Calcium Carbonate and inert fillers shall be at the option of the manufacture, provided all other requirement of MORTH specifications to be met.
Yellow Pigments	-	

2. Properties: The properties of thermoplastic material, when tested in accordance with ASTM D-36 / BS-3262 (Part I) shall be as below:

   a. Luminance:
   White: Daylight luminance at 45º- 65 percent min. as per AASHTO M 249.
   Yellow: Daylight luminance at 45º- 45 percent min. as per AASHTO M 249.

   b. Drying Time:
   When applied at a temperature specified by the manufacturer and to the required thickness, the material shall set to bear traffic in not more than 15 minutes.

   c. Skid resistance: Not less than 45 as per BS 6044.

   d. Cracking resistance at low temperature: The material shall show no cracks on application to concrete blocks.

   e. Softening point: 102.5º± 9.5ºC as per ASTM D 36.

   f. Flow resistance: Not more than 25 per cent as per AASHTO M 249.

   g. Yellowness index (for white thermoplastic paint: not more than 0.12 as per AASHTO M 249.

3. Storage life: The material shall meet the requirements of these specifications for a period of one year. The thermoplastic material must also melt uniformly with no evidence of skins or un-melted particles for the one year storage period. Any material not meeting the above requirements shall be replaced by the Manufacture/Supplier/Contractor.

4. Reflectorisation: Shall be achieved by incorporation of beads, the grading and other properties of the beads shall be as specified in clause 803.4.2 of MORTH specifications.

5. Marking each container of the thermoplastic material shall be clearly and indelibly marked with following information.

   - The name, trade mark or other means of identification of manufacturer.
   - Batch number
   - Date of manufacture.
   - Colour (white or yellow)
   - Maximum application temperature and maximum safe heating temperature.

6. Sampling and Testing: The thermoplastic material shall be sampled and tested in accordance with the appropriate ASTM/BS method. The Contractor shall furnish to the AE a copy of MTC of the thermoplastic material showing results of all tests specified herein and shall certify that the material meets all requirements as mentioned above.

7. Reflective Glass Beads:

   a. General: This specification covers two types of glass beads to be used for the production of reflective pavement marking. Type 1 beads are those which are a constituent of the basic thermoplastic compound vide table 800-3 of MORTH specifications and Type -2 beads are those which are to be sprayed on the marked surface vide clause 803.6.4 of MORTH specifications. The glass beads shall be transparent, colourless and free from milkiness, dark particles and excessive air inclusions. These shall conform to the requirements spelt out in Clause 803.4.2.3 of MORTH specifications.

   b. Specific Requirements:

      - Gradation: The glass beads shall meet gradation requirements for the two types as given in the following table:

Sieve size	Percent Retained (Table 800-10)	
	Type-1	Type-2
1.18 mm	0 to 3	-
850 microns	5 to 20	0 to 5
600 microns	-	5 to 20
425 microns	65 to 95	-
300 microns	-	30 to 75
180 microns	0 to 10	10 to 30
<180 microns	-	0 to 15

      - Roundness: The glass beads shall have a minimum of 70 per cent true spheres.

      - Refractive index: The glass beads shall have a minimum refractive index of 1.50.

      - Free flowing properties: The glass beads shall be free of hard lumps and clusters and shall dispense readily under any condition suitable for paint stripping. The y shall pass the free - Flow test.

2.1.16.4. Traffic Arrangement and Control

1. Before taking up of laying markings, program for the diversion of traffic shall be drawn up and got it approved by the AE as per clause 112 of MORTH specifications and as per IRC SP: 55.

2. Temporary sign boards, barricading, flags, lights and flagmen will be provided at working area as per approval of the AE.

3. On both sides, suitable regulatory/warning signs as approved by the Engineer shall be installed for the guidance or roads users.

2.1.16.5. Work Methodology

1. Equipment's
   a. Pre-Melter
   b. Pram-applicator
   c. Mechanical Broom
   d. Air compressor
   e. Truck
   f. Water tanker with pipe attachment
   g. Survey kit / instrument
   h. Infra-Red Thermometer (Laser thermometer)

2. Application
   a. Preparation (as per MORTH specifications Clause: 803.4.4):

      i. The material shall be melted in accordance with the manufacturer's instructions in a Heater fitted with a mechanical stirrer to give a smooth consistency to the thermoplastic material to avoid local overheating. The temperature of the mass shall be within the range specified by the manufacturer, and shall on no account be allowed to exceed the maximum temperature stated by the manufacturer. The molten material should be used as expeditiously as possible and for thermoplastic material which has natural binders or is otherwise sensitive to prolonged heating the material shall not be maintained in a molten condition for more than 4 hours.

      ii. After transfer to the laying equipment, the material shall be maintained within the temperature range specified by the manufacture for achieving the desired consistency for laying.

         - Cleaning: The surface of the flexible road pavement shall be thoroughly cleaned form all of dust any other material, grease, oil etc. at the area where the material shall be applied to mark the pavement with the help of broom and air compressor.

         - Pre- Marking: The reference centre-line (Pre-Marking) of the lane marking as per the drawing shall be made to have temporary marking with chalk powder/long thin rope ,so that the applicator machine pointer in the front of the machine can be guided to move along this pre-Marked line while material is being applied on the road . This ensures accurate alignment of the markings when laid on the road surface.

         - Heating of the Material: The road marking material shall be heated in the pre-melter as well as pram-applicator machine fitted with mechanical stirrer to give smooth consistency to the material and to avoid local heating of the material. For heating, LPG with suitable burner system shall be used to achieve temperature of 180+/-10 ºC which is the specified application

temperature. The temperature Gauge fitted on the machine as well as the pre-melter displays the material temperatures continuously for monitoring the application temperature.

- Lane Markings: The lane marking shall be made as per the approved drawing for width; length and thickness shall be made with the help of the Screed- Box attached to the Applicator Machine. The thermoplastic paint shall be applied in intermittent of continuous lines of uniform thickens of at least 2.5 mm. Where arrows or letters are to be provided, thermoplastic compound may be hand sprayed. The tolerances of the width, position alignment and length of segment of broken lines shall be as per Technical specification sub clause 803.6.6.

- Drop-On Glass Beads Application: The approved glass beads shall be placed in position from the bead dispenser attached to the machine and the dispersion is by forced-feed mechanism which works while the machine is applying the markings with the Screed-Box., simultaneously and before the thermoplastic material (which is hot applied) becomes dry and cold while achieving road surface temperature. The type-2 glass beads conforming to the above noted specification shall be sprayed uniformly into a mono-layer on to the paint line at the rate of 250 grams per square meter area.

- Weather: The marking shall be done with approved material only on dry pavement surfaces of minimum Temperature of 10º C.

- Arrow Marking: The arrow marking shall be made using the approved material with specifically made stencils of shape and size.

- Properties of finished road marking (as Per MORTH specifications Clause: 803.4.6):

  - The strip shall not be slippery when wet.
  - The marking shall not lift from the pavement in freezing weather.
  - After application and proper drying, the stripe shall show no appreciable deformation or discoloration under traffic and under road temperatures up to 60ºC.
  - The marking shall not deteriorate by contact with sodium chloride or oil drippings from traffic.
  - The stripe or marking shall maintain its original dimension and position. Cold ductility of the material shall be such as to permit normal movement with surface without chopping or cracking.

2.1.17. Methodology for Longitudinal Lined Drain

2.1.17.1. Scope

The Scope of this work covers the following activities:

1. After setting out, the excavation necessary for construction of Longitudinal Drain shall be carried out as per Section 300 MORTH specifications.

2. The selection of various construction materials, testing of materials, approvals and quarry selection, testing and approval of various grades of materials etc. conforming to section 1000 MORTH specifications
3. The construction of Longitudinal Drain with reference to Section 300, Section 1000, Section 1500, Section 1700 of MORTH specifications and IS: 1077.

2.1.17.2. Materials

1. Bricks shall comply Clause 1003 of MORTH specifications.
2. OPC 43 grade conforming to IS 8112 or PPC.
3. Coarse Aggregates, Sand / Fine Aggregates as per IS 383 and the tests shall be carried out as per IS 2386.
4. The formwork material shall comply with IRC 87.
5. Water conforming to section 1010 MORTH specifications.

2.1.17.3. Plant, Equipment and Machinery

The following plant and equipment are necessary for the planned progress:

1. Batching Plant.
2. Transit Mixers.
3. Needle Vibrators.
4. Slump cone and concrete test cube moulds.
5. Concrete Pump and Crane bucket.
6. JCB/Excavator.
7. Tippers/Dumpers.
8. Steel Shutters as formwork.

2.1.17.4. Construction operations.

1. Survey & Setting out Works:
   The limits for excavations shall be set out true to lines, curves, slopes, grades and sections as shown on the drawings and using established bench marks / co-ordinates. All the survey instruments used shall be calibrated periodically as per the Calibration Plan. Survey check reports and other relevant data shall be maintained in relevant inspection formats.

2. Detailed drawing submission:
   The detailed drawing for the longitudinal drain shall be made with the invert levels shown with the indication of type of bedding as specified on the working drawing.

   Before any construction work is being carried out, the Contractor must submit an elevation of the invert level showing the length to be constructed and the following details to be incorporated on the drawing:

   a. The existing ground profile.
   b. Founding levels at the base.
   c. Invert levels of longitudinal drain from one point at downstream to upstream.
   d. The invert levels of the longitudinal drain intersecting the disposal outlet.

   The AE will review for the drawing prior start of the longitudinal drain.

3. Construction of Longitudinal Drain
   a. Excavation:

The foundation bed for the Longitudinal Drain shall be excavated true to the lines and grade as set out. It shall be executed to provide sufficient room for steel shutters.

b. Bedding for Longitudinal Drain:
The bedding surface of filling area must be a firm ground of uniform density throughout the length of the drain. It shall be compacted and tested so that it is good for the PCC to be laid on it.

The bedding surface of cutting area shall be clean and free from loose materials. Then lay the PCC to required thickness and dimensions. The finished surface should be uniform in thickness.

c. Construction of Longitudinal Drain
   i. Clean up any debris in the PCC by blowing out any foreign materials.
   ii. The base slab dimensions shall be marked on the PCC
   iii. All temporary shuttering shall be checked so that they are securely fixed.
   iv. After taking necessary approval from AE pouring of concrete shall be started.
   v. The concrete should be uniformly vibrated with a needle. Conduct the routine slump checks and cast test cubes as per prescribed frequency or as directed by the AE.
   vi. 25mm Expansion joint gap to be provided at every 40 m interval and to be filled with Shalitex board and filler material (water proof).
   vii. Upon completing the concreting it is essential to cure the concrete with potable water on the next day and regularly do so for the minimum next 7 days.

2. 2. Section 2: Structure Works

2.2.1. Methodology for Structural Concrete for Bridges and Culverts

2.2.1.1. Scope

The work shall consist of producing and placing the concrete of grade M15 and higher by weigh batching in accordance with the specifications conforming to MORTH specifications, IRC: 112 and the work specifications as stipulated in the contract document and drawings.

2.2.1.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. IRC: 112
4. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
5. Relevant Drawings

2.2.1.3. Setting Out

After completion of site preparation, alignment of structure is marked on the ground with lime and or string. With reference to its alignment other points of structures like wing wall, abutment wall are pegged. For construction purpose temporary bench marks (TBM) are also established near by the location of structure. The place of TBM shall be chosen in such a way that it will not be disturbed by the construction activities and it shall remain in position until all the works are completed in that location.

2.2.1.4. Material

1. Cement: Cement conforming to IS: 12269 shall be used only after ensuring the minimum required design strength. MTC shall be submitted to the AE for every consignment of cement. The certificate shall cover all the tests for chemical requirements, physical requirements and chloride content. Independent tests of samples drawn from the consignment shall be carried out at the site laboratory or in an independent laboratory approved by the AE immediately after delivery.

2. Coarse Aggregates: Coarse aggregate shall be produced from approved quarry in required sizes. It shall consist of clean, hard, strong, dense, non-porous and durable pieces of crushed stone or a suitable combination thereof as specified. Maximum nominal size of aggregate for various structural components shall conform to the Table 1700-7 of MORTH specifications.

3. Fine Aggregate: Fine aggregate for structural concrete shall be clean, hard, strong and durable pieces of crushed stone or suitable combination of natural sand. They shall not contain dust, lumps, soft or flaky, materials, mica or other deleterious materials in such a quantity to reduce the durability and strength of concrete. Gradation of fine aggregate shall conform to Table 1000-2 of MORTH specifications.

4. Water: Water shall conform to section 1010 of MORTH specifications. Water shall be free from injurious materials like oils, acids, alkaline, salts, sugar, organic materials or other substance that may be deleterious to the concrete or steel. Potable water is generally considered satisfactory for construction purpose. Mixing and curing with seawater shall not be permitted.

5. Reinforcement Steel: Reinforcement steel shall conform to section 1009.3 of MORTH specifications. Reinforcement shall be thermo-mechanically treated (TMT) deformed bars of grade Fe-500 confirming to IS 1786. Only uncoated steel shall be used as per shape and size shown on the drawings. It shall conforms to the specification or as approved by the AE.

6. Bending of Reinforcement: Bar bending schedule will be prepared and submitted to the AE for approval before start of work. Bar bending schedule shall include number, shape and cutting length of bar and weight in respect of each type. A separate bar bending schedule shall be prepared for auxiliary bars like spacer, chairs etc.

All the material shall be got approved from AE prior the utilization in construction works.

2.2.1.5. Placing of Reinforcement

Reinforcement shall be placed in accordance with the drawing and shall be inspected and approved by AE. After getting approval from AE, immediately pouring of concrete can be allowed.

Reinforcement shall be tied with binding wire (1 mm dia.) in such a way that it shouldn't be displaced while pouring the concrete on it.

Cover blocks shall be made of cement, sand & 10 mm down aggregate with the same durability of surrounding concrete and same grade of concrete.

Layers of bar shall be separated by spacer bars kept at approximately 1m interval and min. size of 12mm, or maximum size of main reinforcement, whichever is greater. Horizontal reinforcement shall not be allowed to sag between supports.

Necessary auxiliary bars like spacer, chairs, and blocks shall be provided to fix the reinforcement in its correct position.

2.2.1.6. Proportioning of Concrete

The concreting will be done based on mix design. All measuring equipment shall be in a clean and serviceable condition. Their accuracy shall be periodically checked. Water cement ratio shall be maintained constant to its connecting value taking into account moisture content of aggregates. The amount of water to be added shall be adjusted to compensate the variation in moisture content due to weather conditions. Proportioning of concrete shall be as per approved design mix by weigh batching.

2.2.1.7. Trial Mixes

The trial mixes of all grades of concrete shall be prepared by using samples of approved materials same as that are proposed to use in the works before commencement of work. The initial trial mixes shall be prepared at site laboratory. In all cases complete testing of all the constituent materials proposed in the design mix shall be carried out prior to making trial mixes.
Additional trial mixes shall be carried out before the production, in case of substantial changes are made in the material or in the proportion of the materials to be used.

2.2.1.8. Equipment's

Keeping in view of the requirements of the work, equipment proposed to carry out work is as under:

1. Batching Plant
2. Mixer Trucks / Transit mixer
3. Vibrator
4. Vibrator needles
5. Wheel Loader
6. Excavator
7. Shutter Material
8. Bar Bending & Cutting machine
9. Concrete Pump

2.2.1.9. Transporting, Placing & Compaction of Concrete

Transportation of concrete shall be carried by transit mixer or pumps. Hose pipe from the concrete pump chutes to the placing area should be laid out with minimum bends. Concrete at the time of pouring should have temperature in between 5ºC to 40ºC. It shall be compacted to its final position within 30 minutes of its discharge from the mixer. Concrete shall be poured in horizontal layer of compacted thickness of 300 mm. Fresh concrete shall not be placed against concrete which has been in position for 30 minutes or as fixed by mix design unless appropriate construction joints are provided. Placement of concrete shall not be carried out when the day temperature exceeds 40 ºC.

2.2.1.10. Formwork

Formwork shall conform to section 1500 of MORTH specifications. Material and components used for formwork shall be examined for damage or excessive deterioration before use / reuse and shall be used only if found suitable after necessary repairs.

Only steel formwork shall be used. The steel used for forms shall be of such thickness that the forms remain true to shape. All bolts shall be countersunk. The use of approved internal steel ties or plastic spacers shall be permitted. Structural steel tubes used as support for forms shall have a minimum wall thickness of 4 mm. The design erection & removal of form work shall conform to IRC: 87.

2.2.1.11. Construction Joints

Construction joints shall be provided as minimum as possible and shall be as per drawings. Laitance shall be removed before fresh concrete is cast. The surface shall be roughened enough not to dislodge the coarse aggregate. Concrete may be brushed with stiff brush in its green stage at the location of construction joint. Fully hardened concrete shall be treated with mechanical tools to remove the laitance.

2.2.1.12. Curing

Water curing shall be done with water specified in section 1000 of MORTH specifications. Exposed surface of concrete shall be kept continuously in a damp or wet condition by spraying of water or by covering with a layer of sacks, canvas, hessian or similar materials and shall be kept constantly wet for a period of not less than 14 days from the date of placing the concrete.

2.2.1.13. Finishing

The finishing shall be done as specified in clause 1713 of Section 1700 of MORTH specifications.

2.2.2. Methodology for Box Culverts (Precast/ Cast in Situ)

2.2.2.1. Scope of Work

The work shall consist of reinforcement cement concrete work and furnishing the structure for box culvert at the location shown in the relevant drawing and in accordance with the requirements of the CA. RCC box culverts have been proposed for design and construction of project. The span arrangement is 1m X 1.5m, 1m X 2m, etc. in general or as per reviewed drawings. The construction of RCC box culverts shall be carried out as laid out in relevant IRC code. It is proposed to RCC box culverts consisting of bottom slab, vertical walls and top slab. Cantilever return walls, suitable reinforcement will protrude out of end culvert. Cantilever return walls shall be cast after achieving full box section.

2.2.2.2. Reference

1. Contract Agreement
2. IRC SP: 84-2014
3. Ministry of Road Transport and Highway Specifications (MORTH-5th Revision)
4. Relevant drawings

2.2.2.3. Setting Out

After completion of site preparation, alignment of above said culverts is marked on the ground with lime and /or string. With reference to its alignment other points of structures like wing wall/Return wall, abutment wall are pegged. For construction purpose temporary bench mark (TBM) is also established near by the location of structure. The place of TBM shall be chosen in such a way that it will not be disturbed by the construction activities and it shall remain in position until all the structure are completed in that location.

2.2.2.4. Selection of Material

1. **Formwork:** All materials for formwork shall comply with the requirements of IRC: 87. Only steel formwork shall be used. All bolts shall be countersunk. The use of reviewed internal ties or plastic spacer shall be used.

2. **Reinforcement:** Reinforcement shall be thermo-mechanically treated (TMT) deformed bars of grade Fe-500 confirming to IS code 1786. Only uncoated steel shall be used as per shape and size shown on the drawing. It confirms to the specification or as a reviewed by the Engineer.

3. **Concrete & Brick:** All materials for this concrete shall confirm to the section 1000 of MORTH specifications.

2.2.2.5. Equipment's

The following equipment's shall be used for the construction of Box Culverts.

1. Batching Plant
2. Power Arrangement (Generator / electricity)
3. Mixer Trucks / Transit mixer
4. Concrete Pump
5. Cube Moulds
6. Vibrator
7. Vibrator needles
8. Wheel Loader
9. Excavator
10. Shutter Material
11. Box Segment (If Required)
12. Bar Bending & Cutting machine
13. Hydraulic crane
14. Water Tanker
15. Tractor trolleys
16. Truck Trailer

2.2.2.6. Method of Operation:

1. **Formwork**
   a. **Design of Formwork:** Complete design and drawings of formwork for slab super structure will be submitted for approval before start of work. The design of formwork shall confirm to the IRC 87.

   b. **Workmanship:** Workmanship shall comply with the clause 1504.2, 1504.4 & 1504.10 of MORT&H specifications. Form shall be made so as to produce a finish concrete true to shape, line and levels and dimensions as shown on the drawings, subject to the tolerances specified in respective sections of these specifications, or as directed by the engineer.

   c. All steel tubes are free from kink and bends and it is allowed to rest on firm base. Subsoil condition is also taken into account while designing the staging to avoid on toward failure.

   d. **Preparation of Form before Concreting:** Contact surfaces of the form shall be thoroughly cleaned and dried before applying a release agent. It should be ensured that the release agent is appropriate to the surface to be coated. The form shall be

thinly coated with releasing agent in a sequence first vertical and then horizontal surface and any excess can be wiped out. The same type and make of release agent shall be used throughout or similar formwork materials and different type shouldn't be mixed. The release agent shall not come in contact with the reinforcement bar or the hardened concrete. Release agent should preferably be water based and shall be got reviewed by the AE.

2. **Removal of Formwork of Slab / Super Structure:** The scheme for removal of form (i.e. de-shuttering and de-centring) shall be planned on advance and submitted to the AE for scrutiny and review.

3. **Reinforcement**
   a. **Bending of Reinforcement:** Bar bending schedule shall be prepared and shall be submitted to AE for its review before start of work. Bar schedule shall include number, shape and cutting length of bar and weight in respect of each type. A separate bar bending schedule shall be prepared for auxiliary bars like spacer, chairs etc.
   b. **Placing of Reinforcement:**
      i. Reinforcement shall be placed in accordance with the drawing and get inspected and reviewed by the AE. After reviewed from AE, immediately pouring of concrete follows.
      ii. Reinforcement shall be tied with binding wire in such a way that it won't be displaced while pouring the concrete on it.
      iii. Cover block are made of cement, sand and 10mm down aggregate with the same durability of surrounding concrete.
      iv. Layers of bar shall be separated by space bars kept at approximately 1m interval with min size of 12 mm and max size of main reinforcement, whichever is greater is used. Horizontal reinforcement shall not be allowed to sag between supports.
      v. Necessary auxiliary bars like spacer, chairs and blocks shall be provided to fix in its position as per standard.

4. **Concrete**
   a. **Grades of Concrete:** Concrete used in any structure shall be specified in reviewed Drawings. For all items of work Design Mix shall be used.

   b. **Proportioning of Concrete:** Proportioning of concrete shall be done in computerized batching plant. All measuring equipment shall be in a clean and serviceable condition. Their accuracy shall be periodically checked. Water cement ratio is maintained constant to its designated value taking in to account of moisture content of aggregates. The amount of water to be added shall be adjusted to compensate for the variation in moisture content due to weather condition.

   c. **Admixtures:** Duly tested admixtures/ additives confirming to IS 6925 and IS 9103 (without replacement of cement) may be used subject to satisfactory proven use, with the approval of the Authority Engineer. Admixture generating Hydrogen or Nitrogen and containing chloride, sulphide, sulphates, nitrates and any other materials liable to affect steel or concrete shall not be used.

   d. **Size of Coarse Aggregate:** Size of coarse aggregate shall refer to the in MORT&H Specification table no. 1700-7.

5. **Transporting, Placing and Compaction of Concrete:**
   Transportation of concrete is carried by transit mixer. Placing of concrete should be by using suitable chutes/concrete pump/ as per site requirement. It shall be compacted to its final position within 30 minutes of its discharge from the mixer. Concrete shall be poured in horizontal layer of compacted thickness of 300 mm. Fresh concrete shall not be placed against concrete which has been in position for 30 minutes unless appropriate construction joins is provided. The Sequence of concrete shall be as follows:
   a. Soffit Slab
   b. Wall
   c. Top Slab
   d. Return wall

   The concrete of design grade with a slump of 80mm to 120mm shall be produced at the centralized batching plant, and transported to location by transit mixers. The concrete shall be placed in by concrete pump. The chute/flexible pipe at the end of the pipe line shall be positioned to allow direct pour of concrete. Concrete buckets shall be arranged as back up. The concrete shall be vibrated by needle vibrators (40mm / 60mm dia needles). Cubes in required number shall be cast from respective concrete pours.

6. **Protection and Curing:** Weep Hole shall be provided as per relevant drawing of the structure.

7. **Protection and Curing:**
   Exposed surfaces of concrete shall be kept continuously in a damp or wet condition by ponding or by covering a wet layer of sacks, canvas, Hessian or similar materials and shall be kept continuously wet for a period of 14 days from the date of placing of concrete.

8. **Backfilling:**
   Approved materials in layers not exceeding 200 mm. Backfill shall be placed in uniform layers along the sides of the boxes and over the top of the box sections wherever required. Backfilling shall be carried out in uniform layers along the sides of the boxes and over the top of the box sections wherever required.

2.2.3. Methodology for Pipe Culverts

2.2.3.1. Scope

The work shall consist of furnishing and installing required length, type and size of reinforced cement concrete pipe at the location shown in the drawing or in accordance with the requirements of the CA.

2.2.3.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.2.3.3. Setting Out

After completion of site clearance, alignment of culvert is surveyed to trace out lines on the ground with lime and or string for the location of culvert and its structures. For construction purpose temporary bench mark (TBM) is also established near by the location of structure. The

place of TBM shall be chosen in such a way that it will not be disturbed by the construction activities and it should remain in position until all the structure are completed in that location.

2.2.3.4. Working Drawings

Working drawings, structural drawings along with relevant details will be submitted to the AE prior to start of work on site. After review of the above submissions by the AE, the setting out of the structures shall be done.

2.2.3.5. Selection of Material

All materials used in the construction of pipe culverts shall conform to the requirements of Section 2900 of MORTH specifications. Concrete pipes to be used in site, shall be inspected, tested, if necessary and approved by the AE and provided with manufacturing certificates.

2.2.3.6. Equipment's

The following equipment's shall be employed for the construction

1. Hydraulic Excavator
2. Hydraulic crane / Hydra
3. Dumper
4. Plate compacter, mechanical or pneumatics tamping
5. Water Tanker
6. Tractor trolleys

2.2.3.7. Method of Operation

1. The foundation bed for pipe culverts shall be excavated true to the lines and grades shown on the drawing or as directed by the AE. The pipe shall be placed in shallow excavation of the natural ground or in open trenches cut in existing embankment, taken down to levels as shown on the drawings. Where trenching is involved it's width on either side of the pipe shall be a minimum of 150 mm or one-fourth of the diameter of the pipe whichever is more. But in no case it should be more than one third of the diameter of the pipe. The sides of the trench are kept as nearly vertical as possible. Pipe shall be placed on PCC bed.

2. The bedding surface is provided with firm foundation of uniform density throughout the length of the culvert in specified line, levels and grades.

3. After completing foundation, then only pipe is laid in position in required levels and grades without damage as per approval from the AE. If many pieces are to be laid, then clear spacing is maintained at least half the diameter of the pipe to a minimum of 500mm. Placing of pipe is carried out beginning from outlet (down slope) towards inlet (upslope) in a grade to make invert smooth as specified in drawing. In the case of bell-mouth pipe, belled end of the pipe should face upstream.

4. The pipes are joined as per the connection details show in the approved drawings. For joining pipe under light hydraulic pressure the recess at the end of the pipe shall be filled with jute braiding dipped in hot bitumen or other suitable approved compound. The bitumen ring of one pipe shall set in to recess of the next pipe. The ring shall be thoroughly compressed by jacking or by any other suitable methods. All joints should consistent with the interior surface with the pipes. After finishing, the joints shall be kept covered and damp for at least four days.

5. Backfilling of trenches shall commence immediately after jointing material has hardened. The backfill soils shall be clean, free from boulders, large roots excessive amounts of sods or other vegetables matter, and lumps and shall be approved by the AE. Filling of trench is carried out on both side of the pipe to avoid uneven pressure to the pipe. Backfilling are carried out in layer of not exceeding 150mm, and is thoroughly rammed, tamped and vibrated in each layers. Approved pneumatic or light mechanical tamping equipment can be used. Special attention shall be paid on the hunch of the pipe to consolidate the filling materials.

6. Head walls, wing walls, aprons and other ancillary works shall be constructed in accordance with the details as shown in the drawings. RCC wall shall confirm to the section 1500, 1700, 2100, 2200 or 2300 as applicable Aprons shall confirm to the section 2500.

2.2.3.8. Opening to Traffic

To open the traffic through the culvert, minimum earth cushion required above the top of the pipe is atleast 600 mm.

2.2.4. Methodology for Slab Culverts and Minor Bridges

2.2.4.1. Scope

The work shall consist of reinforcement cement concrete work and furnishing the structure for slab culverts and minor bridges at the location shown in the drawing or as instructed by the Engineer and in accordance with the requirements of the MORTH specifications.

2.2.4.2. Reference

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings

2.2.4.3. Setting Out

After completion of site preparation, alignment of said culverts or minor bridge shall be marked on the ground with lime or string. With reference to its alignment other points of structures like wing wall / return wall, abutment wall are pegged. For construction purpose temporary bench mark (TBM) is also established near by the location of structure. The place of TBM shall be chosen in such a way that it will not be disturbed by the construction activities and it shall remain in position until all the structure are completed in that location.

2.2.4.4. Working Drawings

Working drawings, structural drawings along with relevant details will be submitted to the AE prior to start of work on site. After review of the above submissions by the AE, the setting out of the structures shall be done.

2.2.4.5. Selection of Material

1. Formwork: All materials for formwork shall comply with the requirements of IRC: 87. Only steel formwork shall be used. All bolts shall be countersunk. The use of approved internal ties or plastic spacer shall be used. Structural steel tubes used as support for forms shall have a minimum wall thickness of 4mm.

2. Reinforcement: Reinforcement shall be thermo-mechanically treated (TMT) deformed bars of grade Fe-500 confirming to IS 1786. Only uncoated steel shall be used as per shape and size shown on the drawing. It shall confirm to the specification or as approved by the AE.

3. Concrete & Brick: All materials for this work i.e. concrete / brick shall confirm to the section 1000 of MORTH specifications.

2.2.4.6. Equipment's

The following equipment's shall be used for the construction of Minor Bridge and Culverts:

1. Batching Plant
2. Mixer Trucks / Transit mixer
3. Vibrator
4. Vibrator needles
5. Wheel Loader
6. Excavator
7. Shutter Material
8. Bar Bending & Cutting machine
9. Concrete Pump
10. Hydraulic crane
11. Water Tanker
12. Tractor trolleys

2.2.4.7. Method of Operation

1. Formwork
   a. Design of Formwork: Complete design and drawings of formwork for slab super structure will be submitted for approval before start of work. The design of formwork shall confirm to the IRC 87.

   b. Workmanship: Workmanship shall comply with the clause 1504.2, 1504.4 & 1504.10 of MORTH specifications. Form shall be made so as to produce a finish concrete true to shape, line and levels and dimensions as shown on the drawings, subject to the tolerances specified in respective sections of these specifications, or as directed by the AE. All steel tubes are free from kink and bends and it is allowed to rest on firm base. Subsoil condition is also taken into account while designing the staging to avoid onto-ward failure.

   c. Preparation of Form before Concreting: Contact surfaces of the form shall be thoroughly cleaned and dried before applying a release agent. It should be ensured that the release agent is appropriate to the surface to be coated. The form shall be thinly coated with releasing agent in a sequence first vertical and then horizontal surface and any excess can be wiped out. The same type and make of release agent shall be used throughout or similar formwork materials and different type shouldn't be mixed. The release agent shall not come in contact with the reinforcement bar or the hardened concrete. Release agent should preferably be water based and shall be got approved by AE.

   d. Removal of Formwork of Slab/ Super Structure: The scheme for removal of form (i.e. de-shuttering and de-centring) shall be planned on advance and submitted to the AE for approval. No formwork shall be removed without prior approval of AE.

2. Reinforcement
   a. Bending of Reinforcement: Bar bending schedule shall be prepared and submitted to AE for approval before start of work. Bar bending schedule shall include number, shape and cutting length of bar and weight in respect of each type. A separate bar bending schedule shall be prepared for auxiliary bars like spacer, chairs etc.

   b. Placing of Reinforcement:
      i. Reinforcement shall be placed in accordance with the approved drawings and get inspected and approved by the AE. After approval pouring of concrete shall follow.
      ii. Reinforcement shall be tied with binding wire in such a way that it won't be displaced while pouring the concrete on it.
      iii. Cover block are made of cement, sand and aggregate with the same durability of surrounding concrete.
      iv. Layers of bar shall be separated by space bars kept at approximately 1m interval and min. size 12mm or maximum size of main reinforcement, whichever is greater is used. Horizontal reinforcement shall not be allowed to sag between supports.
      v. Necessary auxiliary bars like spacer, chairs, and blocks shall be provided to fix in its position.

2.2.4.8. Concrete

1. Grades of Concrete: Concrete used in any structure shall be specified by designation. For all items of work Design Mix shall be used.

2. Proportioning of Concrete: Proportioning of concrete shall be done in computerized batching plant. All measuring equipment shall be in a clean and serviceable condition.

3. Their accuracy shall be periodically checked. Water cement ratio is maintained constant to its designated value taking in to account of moisture content of aggregates. The amount of water to be added shall be adjusted to compensate for the variation in moisture content due to weather condition.

4. Admixtures: Duly tested admixtures/ additives confirming to IS 6925 and IS 9103 (without replacement of cement) may be used subject to satisfactory proven use, with the approval of AE. Admixture generating Hydrogen or Nitrogen and containing chloride, sulphide, sulphates, nitrates and any other materials liable to affect steel or concrete and shall not be used.

5. Size of Coarse Aggregate: Size shall refer to Table 1700-7 of MORTH specifications.

2.2.4.9. Transporting, Placing and Compaction of Concrete

Transportation of concrete shall be carried out with transit mixer. Placing of concrete shall be done by using suitable chutes or concrete pump. Pipe lines from the pump and to the placing area shall be laid out with minimum bends. Concrete at the time of pouring should have temperature in between 5 ºC to 40ºC. It shall be compacted to its final position within 30 minutes of its discharge from the mixer. Concrete shall be poured in horizontal layer of compacted thickness of 300 mm. Fresh concrete shall not be placed against concrete which has been in position for 30 minutes unless appropriate construction joins is provided. Placement of concrete shall not be carried out when the day temperature exceeds 40 ºC.

2.2.4.10. Protection and Curing

Exposed surfaces of concrete shall be kept continuously in a damp or wet condition by ponding or by covering a wet layer of sacks, canvas, hessian or similar materials and shall be kept continuously wet for a period of 14 days from the date of placing of concrete.

2.2.5. Methodology for Bridge Pile Foundation

2.2.5.1. Scope

This method statement covers the boring of pile for new if any Minor Bridges / ROB in accordance with the details shown on the relevant drawings and conforming to the requirement of MORTH specifications. The work shall be carried out as per IS: 2911 Part 1.

2.2.5.2. References

1. Contract Agreement
2. IRC: SP: 84-2014
3. Ministry of Road Transport& Highway Specifications (MORTH -5th Revision)
4. Relevant Drawings
5. IS: 2911 Part1

2.2.5.3. Setting Out

1. The centre line / bench mark of each bridge / structure shall be fixed with reference to reference point / project centre line & get it approved by the AE. The reference points to be fixed to mark X-X axis (usually in traffic direction) & Y-Y direction. For the establishment of above points Total Station may be used.

2. For every bridge / structure at least 2 nos. of permanent bench mark / centre line points shall be fixed on either side of bridge keeping a safe distance from the construction area so that in future it should not be disturbed by any means.

3. Centre line / bench mark should be transferred to every pier / abutment location keeping a safe distance from foundation area in X-X and Y-Y direction.

4. The layout with help of co-ordinates of all Piles coming under the particular pier/abutment to be made before starting of boring any Pile.

5. These working points should be checked along with AE and shall be entered in the site record.

2.2.5.4. Working Drawings

Working drawings, structural drawings along with relevant details will be submitted to the AE prior to start of work on site. After review of the above submissions by the AE, the setting out of the structures shall be done. Test Pile: Test Pile shall be done as per the reviewed working drawings and clauses 1105 of MORTH specifications.

2.2.5.5. Material

The basic material shall conform to section 1000 of MORTH specifications. The structural steel, concrete shall be used in pile foundations shall conform to Section 1600,1700,1800,1900 of MORTH specifications.

2.2.5.6. Equipment's

The following equipment's shall be deployed in the execution of Piling at Minor bridges.

1. Piling Rig / Percussion machine and other allied equipment's.
2. Lifting Crane
3. Dumpers
4. Batching & Mixing Plant
5. Transit Mixers
6. Concrete Pump
7. Vibrators
8. Vibrator needles
9. Bar Bending Machine

2.2.5.7. Personnel for Execution & Supervision

A well trained team who has experience to execute such activity shall be deputed to execute the above job. Cage inspections shall be done by the Sr. Structural Engineer / DPM to ensure the conformance to the working drawings and relevant codes before lowering the cage.

2.2.5.8. Construction Operations

1. Boring
   a. The cast-in-situ bored pile shall be done by making bore in the ground and removing the material.
   b. Boring shall be carried out using rotary equipment.
   c. The diameter of the finished pile shall not be less than that specified. A continues record shall be kept as to the volume of concrete placed in relation to the length of pile that is cast.
   d. The drilling mud such as bentonite suspension shall be maintained at a level sufficiently above the surrounding ground water level throughput the boring process, to ensure the stability of the strata which is being penetrated until the pile has been concreted.

2. Concerting
   a. Wherever practicable, concrete should be placed in a clean dry hole.
   b. Before concreting of the pile is commenced, it is essential to ensure that no debris remains at the bottom of the shaft.
   c. Reinforcement for the pile as shown on the reviewed drawing shall be tied in place to form a cage which is lowered into the pile shaft.
   d. Where concrete is placed in dry and a casing is present, the top 3m of the pile shall be compacted using internal vibratos.
   e. Care shall be taken during concerting to prevent the segregation of the ingredient.
   f. The concrete shall be properly graded, shall be self-compacting and shall not get mixed with soil, excess water, or other extraneous matter.
   g. The pacing of concrete with tremie pipe shall be continues process from the toe level to the top of the pile. To ensure compaction by hydraulic static heads, rate of placing concrete in the pile shaft shall not be less than 6m (length of pile) per hour.
   h. It shall be ensured that bottom of tremie pipe always remains buried in already placed concrete.
   i. Temporary Casing during the concerting shall be done as per the clause 1107.3 of MORTH specifications.

2.2.5.9. Mixing, Transportation, placing and compaction of Concrete

1. Concrete mixing: Concrete shall be mixed in approved proportions in batching and mixing plant (clause 1708 of MORTH specifications).

   Batching plant calibration shall be approved by the AE. Mixing time shall be adjusted according to the efficiency of the mixer to ensure the uniform distribution and a uniform colour of the entire batch. In any case mixing time shall not exceed two minutes. Mixer which has been out of use for more than 30 minutes shall be thoroughly cleaned before putting in a new batch. Mixer shall be thoroughly cleaned before switching over from one type of cement to the other type. Water cement ratio of the mix shall be maintained to the designed one by recording the moisture content of the ingredients frequently and adjusting the water quantity of the mix. Suitable adjustment shall also be made in the weight of aggregate to allow for the variation in the weight of aggregate due to variation in their moisture content. Aggregate shall be kept moist or shaded during summer to keep the temperature low. Sand / fine aggregate shall conform to Table 1000-2 of MORTH specifications. Approved super plasticizers shall be used to delay the setting time and increase fluidity of the concrete mix to aid the pumping and long distances transportation. It shall conform to section 1012 of MORTH specifications.

2. Transportation of concrete mix: Concrete shall be transported in transit mixers. Concrete shall not be poured during the peak temperature hours in summer season to keep the maximum temperature below 40ºC. In winter concrete shall be avoided during peak low temperature hours to maintain the minimum temperature more than 5ºC. As per Technical Specifications concreting will not be done when ambient temperature exceeds 40ºC.

3. Placing and compaction of concrete: Concrete shall be poured in horizontal layer of compacted thickness of 300 mm. Fresh concrete shall not be placed against concrete which has been in position for 30 minutes or as fixed by mix design unless appropriate construction joints are provided. Concrete shall be placed as near as possible to its final position so that no contamination, segregation, loss of its constituents material take place. It shall not be freely dropped into place from a height exceeding 1.5 m. All formwork and reinforcement fixed shall be cleaned and made free from standing water and other unsuitable material, prior to concreting.

2.2.5.10. Quality Control

Tolerances shall be as per Clause 1116.1 of MORTH specifications.

2.2.6. Methodology for Pile Cap

2.2.6.1. Survey

Before commencing excavation, the pile cap area shall be marked on the ground after carrying out survey with reference to control points. After excavation the levels of the pit shall be checked for correctness to the drawings and recorded. Longitudinal and transverse centrelines shall be marked outside the pit for reference for cross checking the pile. The shift of pile shall be recorded by marking the theoretical coordinates of pile on ground and circle of equivalent to pile diameter shall be drawn and shift of actual pile will be measured from theoretical edges of pile. The as built details of the piles shall be recorded jointly. Based on shifted location of pile top, pile cap / PCC layout shall be marked to ensure minimum 150 mm offset from outer edge

of pile. After laying of PCC, the layout of the pile cap shall be marked on it with reference to the reference points to facilitate tying of rebar and erection of shuttering.

2.2.6.2. Excavation

Shoring shall be provided depending upon the stability of the soil found in the area. Shoring shall be done with the help of old steel plates and props. At road locations the pit shall be excavated to the dimensions providing working space all around the pile cap, to facilitate fixing of steel& erection of shuttering as detailed in the drawings. The last 300 mm excavation shall be carried out manually & levelling course shall be laid down within 36 hrs after completing excavation of last 200 mm depth. Provision for sump shall be made at the corner of the pit to pump out underground water of about 750 mm deep from PCC bottom. Also an earthen drain of about 200 mm width if required shall be provided all around pile cap dimension & it shall be connected with sump to drain off excess rain water / seepage water to ensure that the water table will at least 300 mm below the lowest level of the excavation before laying PCC. The excavated earth shall be then disposed by means of trucks / loaders.

2.2.6.3. Removal of Laitance

After excavation the laitance of the piles shall be removed by using Pneumatic Jack Hammers seven days after casting of pile or manually three days after casting of pile. The top of pile after striping shall project 50 mm into the pile cap and reinforcements of pile shall be fully anchored in pile cap as per clause 709.5.2 of IRC: 78. The debris of broken concrete shall be removed from the pit and disposed-off to approve dumping sites. Exposed bars shall be straightened & cleaned properly with wire brush.

2.2.6.4. PCC

After levelling the bottom, the pit shall be watered to keep the soil moist. PCC of M15 mix or as specified in drawing shall be mixed at the centralized batching plant at the casting yard and transported to site in transit mixers.

The concrete shall directly pour through chutes, shall be spread and levelled manually to the specified thickness shown in the drawing. PCC edges shall be projected 100 mm more than that of pile cap as detailed in the drawing to facilitate the fixing of form work and levels of PCC shall be jointly checked & verified. The PCC shall be cured by sprinkling water.

2.2.6.5. Reinforcement

1. Fabrication: For fabrication of reinforcement, BBS shall be prepared as per approved drawings. The reinforcement shall be cut using cutting machines or manually as required and bent at rebar yard. The re-bars shall either be transported to location in trailer / truck or cutting bending may be done at site depending upon the situation.

2. Fixing of Re-bars: The re-bars shall be manually fixed into its position as shown in the drawings. Reinforcement of pile cap shall be lapped with the reinforcement of pile. Cover blocks of same grade of concrete in which these are to be embedded shall be provided at spacing of about 2.0 m c/c to ensure uniform cover as specified in the drawing and tied together with GI binding wire. After fixing the pile cap rebar, pier shaft rebar shall be erected. This rebar shall be supported by erecting a suitable staging frame across the width of the pile cap. The rebar cage shall be checked as per checklist given in IRC SP 112 and RFI for inspection shall be raised for inspection of AE. The rebar, chairs, spacers & laps shall be jointly checked after completion of cage placing. Sufficient chairs and spacers shall be provided to keep the cage in its proper position.

After getting the clearance, balance shuttering work will be taken up. Pier dowels shall be hold in position rigidly to prevent it from buckling.

2.2.6.6. Shuttering

Shuttering fabricated as per approved drawings shall be placed at locations as per the pile cap dimensions shown in the good for construction drawing. Reinforcement shall be fixed as per the drawing and marking the layout on PCC for pile cap. After completing the fixing of rebar cage shuttering shall be erected & fixed on the layout drawn on PCC. Before fixing, the shuttering area shall be cleaned with wire brush & approved shuttering oil shall be applied on concrete face. After fixing of shuttering it shall be checked as per QA checklist and RFI to be raised for inspection of AE. For preventing leakages from joints, rubber strip / foam strip shall be provided at the joints of shuttering plates. Proper side supports/bracings/tie bars shall be provided to resist lateral pressure of green concrete during pouring.

2.2.6.7. Concreting

The concrete of required grade shall be produced as per the approved design mix at the centralized batching plant and transported by transit mixers to the pouring location. Before pouring concrete, slump of 80 mm to 130 mm shall be checked at pouring location. The concrete shall be placed by concrete pump / placer boom or by direct chutes. Concrete placing commences from one end to another in cascading manner till completion. The drop height of the concrete should not be more than 1.5 m. The concrete shall be vibrated using 60 mm / 40 mm diameter needle vibrators. Concrete cubes shall be taken for testing of compressive strength as per IRC 21 at pouring location. Concreting should be done in such a way that next layer of concrete should be laid before initial set of concrete of previous layer to avoid cold joints. Regular tamping shall be done during & after concreting operation.

2.2.6.8. Curing

Date of concrete shall be written on the concrete surface to ensure curing up to the specified duration after date of casting. The concrete shall be cured by ponding method. Bunds of cement mortar of lean mix shall be built after the concrete attains final setting time after casting of concrete, these bunds shall be filled with water from approved source. The sides of the pile cap shall be covered with Hessian cloth till back filling is started. Further curing is ensured by keeping the backfill moist with water.

2.2.6.9. Backfilling

Immediately after de-shuttering, concrete surface shall be checked jointly and get the approval for backfilling. Back filling with local earth available or excavated material shall be carried out in layers of 150 mm (compacted thickness) (as per clause 304.3.7 of MORTH specifications). The compaction shall be done with the help of suitable equipment such as mech. tamper, rammer, plate vibrator etc., after necessary watering, so as to achieve a density not less than the field density before excavation.

2.2.6.10. Miscellaneous Works

After backfilling the damaged roads, if any shall be repaired immediately preferably within 15 days of backfilling as per specification before removing of the barricades, the whole area shall be cleaned after completion of work.

2.2.6.11. Construction Machinery

1. Excavator
2. Dumper
3. Transit Mixers

4. Batching Plant
5. Truck / trailer
6. Welding Generator
7. Concrete Vibrators
8. Concrete Needles
9. Jack hammer
10. Compressor
11. Crane / Hydra
12. De-watering Pump

2.2.7. Methodology for Casting of Pier / Abutment Shaft

2.2.7.1. Work Procedure

1. Survey: The location of piers / abutment shafts and their layout shall be checked once again with respect to the reference points already established during pile foundation works of respective pier.

2. Reinforcements: The BBS shall be prepared as per approved drawings and the same shall be submitted to AE. The bar length shall be cut as per length given in submitted BBS either manually or by cutting machine and then those shall be bend as per requirement shown in BBS. The cutting and bending operation shall be done in reinforcement yard or at site depending upon the case. The reinforcement shall be properly cleaned before use. After that reinforcement shall be placed and fixed in position manually or the prefabricated cage can be transported & fixed to reduce the time cycle.

3. Formwork: The form work shall be fabricated in-line with the reviewed drawings in such way to make standards shuttering and if required special shuttering shall be fabricated. All the joints shall be connected by bolts and rubber / foam strips shall be used in all joints to prevent leakage during concreting. Lateral supports/ties shall be provided at suitable spacing to hold the shutters in position while concreting to counter act the lateral pressure due to green concrete. Gaps / holes in the form work shall be plugged with foam / putty or suitable material to make it leak proof.

   The surface of all formwork / shutter which shall receive green concrete shall be thoroughly cleaned and shutter oiled before fixing in position. For adjusting verticality, anchoring shall be made by using wire ropes from top of shutters to suitable anchor points on and around ground surface or concrete block support. Turn buckle / chain pulley shall be attached to the above wire ropes to pull the formwork for ensuring its verticality. Once the formwork is fixed, the same shall be got checked as per checklist format given in IRC SP: 112 and RFI shall be raised for checking / inspection by AE.

4. Concreting: The concrete of required strength shall be prepared at batching plant as per approved mix design in the site laboratory. The concrete shall be transported in transit mixers to the pouring location. Slump in concrete shall be kept between 80 mm to 130 mm at pouring location and the slump shall be checked regularly at the pouring location. The concreting shall be done using concrete pump with flexible hose or with crane & bucket arrangement and free fall of concrete shall be limited to 1.50 m. The compaction of concrete shall be done using needle type vibrators with 60 mm / 40 mm diameter needles. To check the quality of concrete, cubes shall be taken for testing

compressive strength as per clause 1718 of MORTH specifications at pouring locations. Shear key of suitable size shall be provided at horizontal construction joint at the inter face of older lift and new lift of concrete. Also top surface of substructure lift shall be hacked/roughened to remove laitance for proper bonding between two lifts.

5. De-shuttering: De-shuttering will be done after attaining desired strength of the structure and shall be done by using the crane. The joints of shuttering to be loosened and bolts are too opened one by one to take shutter out without damaging the pier ribs.

6. Curing: The concrete shall be cured by putting hessian cloth over the entire surface of pier shaft except the top surface of pier. The hessian cloth shall be kept moist by sprinkling water and for pier top surface water shall be poured directly on the surface to keep it wet throughout the curing period of 14 days. Curing compound may also be used, depending upon the necessity.

2.2.7.2. Deployment of Equipment

1. Batching plant
2. Transit mixer
3. Concrete pump
4. Welding generator
5. Concrete vibrator
6. Vibrator needles
7. Crane
8. Truck/Trailer
9. Bar Bending Machine

The capacity and quantity of above equipment shall be as per the quantum of work as shall be decided by the Site team during the execution of such works at site.

2.2.7.3. Safety

All safety precautions shall be taken as per relevant IRC Specifications. Further the preparatory works shall also be carried out as per diversion plans separately submitted for taking up construction activities in urban areas.

2.2.8. Methodology for Casting of Pier/Abutment Cap

2.2.8.1. Work Procedure

1. Survey: Centre line on both directions shall be marked at the top of the pier shafts for placing formwork for pier cap. After casting of pier shaft, the centreline shall be marked over the top surface of pier to fix the formwork, reinforcement and concreting.

2. Reinforcements: The BBS shall be prepared as per approved drawings and the same shall be submitted to AE. The bar length shall be cut as per length given in submitted BBS either manually or by cutting machine and then those shall be bend as per requirement shown in BBS. The cutting and bending operation shall be done in reinforcement yard or at site depending upon the case. The reinforcement shall be properly cleaned before use. After that reinforcement shall be placed and fixed in position manually or the prefabricated cage can be transported & fixed to reduce the time cycle.

3. Formwork: The form work shall be fabricated in-line with the reviewed drawings in such way to make standards shuttering and if required special shuttering shall be fabricated. All the joints shall be connected by bolts and rubber / foam strips shall be used in all joints to prevent leakage during concreting. Lateral supports/ties shall be provided at suitable spacing to hold the shutters in position while concreting to counter act the lateral pressure due to green concrete. Gaps / holes in the form work shall be plugged with foam / putty or suitable material to make it leak proof.

   The surface of all formwork / shutter which shall receive green concrete shall be thoroughly cleaned and shutter oiled before fixing in position. For adjusting verticality, anchoring shall be made by using wire ropes from top of shutters to suitable anchor points on and around ground surface or concrete block support. Turn buckle / chain pulley shall be attached to the above wire ropes to pull the formwork for ensuring its verticality. Once the formwork is fixed, the same shall be got checked as per checklist format given in IRC SP: 112 and RFI shall be raised for checking / inspection by AE.

4. Concreting: The concrete of required strength shall be prepared at batching plant as per approved mix design in the site laboratory. The concrete shall be transported in transit mixers to the pouring location. Slump in concrete shall be kept between 80 mm to 130 mm at pouring location and the slump shall be checked regularly at the pouring location. The concreting shall be done using concrete pump with flexible hose or with crane & bucket arrangement and free fall of concrete shall be limited to 1.50 m. The compaction of concrete shall be done using needle type vibrators with 60 mm / 40 mm diameter needles. To check the quality of concrete, cubes shall be taken for testing compressive strength as per clause 1718 of MORTH specification at pouring locations. Shear key of suitable size shall be provided at horizontal construction joint at the inter face of older lift and new lift of concrete. Also top surface of substructure lift shall be hacked/roughened to remove laitance for proper bonding between two lifts.

5. De-shuttering: De-shuttering will be done after attaining desired strength of the structure and shall be done by using the crane. The joints of shuttering to be loosened and bolts are too opened one by one to take shutter out without damaging the pier ribs.

6. Curing: The concrete shall be cured by putting hessian cloth over the entire surface of pier shaft except the top surface of pier. The hessian cloth shall be kept moist by sprinkling water and for pier top surface water shall be poured directly on the surface to keep it wet throughout the curing period of 14 days. Curing compound may also be used, depending upon the necessity.

2.2.8.2. Deployment of Equipment

1. Batching plant
2. Transit mixer
3. Concrete pump
4. Welding generator
5. Concrete vibrator
6. Vibrator needles
7. Crane
8. Truck/Trailer
9. Bar Bending Machine

The capacity and quantity of above equipment shall be as per the quantum of work as shall be decided by the Site team during the execution of such works at site.

2.2.8.3. Safety

All safety precautions shall be taken as per relevant IRC Specifications. Further the preparatory works shall also be carried out as per diversion plans separately submitted for taking up construction activities in urban areas.

2.2.9. Bridge Concrete Superstructure

2.2.9.1. Scope

This section shall cover furnishing and providing of concrete superstructure in accordance with the drawings and as per MORTH specifications.

2.2.9.2. Material

Material shall conform to Section 1000 of MORTH specifications.

2.2.9.3. Casting, handling of precast RCC T Girder

The main activities and procedure to be followed for the casting and handling of the pre-cast girders shall be as below:

1. Fixation of formwork: Firm casting beds for girders shall be prepared at the site and side shutters as per dimensions shown in the reviewed drawings shall be placed in position and de-shuttering shall be done after concreting of girders by removing side shutters. Suitable distances have been kept for the side shuttering so that the girders can be removed from the bed.

2. Preparation & Placing of Reinforcement Cage: A standard BBS shall be prepared as per the length of the girders. The cutting, bending & binding of the reinforcement shall be done as per BBS on the platform. The details of the reinforcement placed shall be checked in accordance with the BBS and recorded as per format. The cover blocks provided shall be of the approved shape, size and of the same strength as that of the girders as mentioned in the approved drawing.

3. Pouring of concrete: Concrete of required grade shall be mixed at the batching plant as per the approved mix design, transported to the site by transit mixer, and poured with the help of chute attached to TM concrete shall be poured up to the full height and shall proceed from one end to the other. Care to be taken that no cold joint is allowed to occur during concreting. Compaction shall be ensured by using needle vibrators of size 40 mm and 60 mm as required. Wooden mallets shall be continuously used for tamping on shutters to remove any air bubbles.

2.2.9.4. Identification precast RCC T-Girder

The pre-cast girders shall be marked suitably for identification. These shall have the following markings on each side of the girder:

1. Date of Casting.
2. Length of Girder
3. Girder No.

2.2.9.5. Curing and de-position of RCC I Girders from Bed

The curing of RCC T Girder shall be done as prescribed in MORTH specifications. The RCC I Girders shall be shifted from the bed after 28 days only.

2.2.9.6. Tolerance

Tolerance shall be as per MORTH specifications Clause 2307.1.

Variation in thickness of top and bottom flange for T-girders and slabs.	±5 mm
Variations in web thickness	-5 mm to +10 mm
Variation in overall depth or width	±5 mm
Variation in length overall and length between bearings	Shall not exceed ±10 mm or ±0.1% of the span length, whichever is less.
Permissible surface unevenness in deck slab when measured with a 3 m straight edge or template.	5mm

2.2.9.7. Pedestals Casting & Bearing Fixing

1. The centre line of bearing should be marked on the pier/abutment cap by the surveyor and size of pedestal shall be marked accordingly.
2. The size of pedestals over pier/abutment cap shall be cast as per the reviewed drawings.
3. Around bearing there shall be a clear offset of 150 mm beyond the edges of bearings or the size of pedestals shall be so kept so that bursting/crushing of pedestal does not occur under the load coming on it.
4. The shuttering shall be erected as per required size of pedestal.
5. For installation of bearing, pockets commensurate with the size of anchors shall be kept in pedestals during concreting of the same.
6. The pedestal shall be cast approximately 20mm short of the required finished level so that the required margin for fixing of bearing is available, after that gap shall be filled with the cement epoxy mortar.

2.2.9.8. Launching of Girders

1. RCC T Girders shall be lifted by lifting hook provided in the girder on both ends with the help of two cranes. Both cranes have to act simultaneously and slowly and girder shall be placed directly on already placed bearing as per the drawings.
2. The girders shall be placed in its true alignment & position and in plumb with the help of Sims and shall be checked for their location, level so that they are as per drawings.
3. Where the gap between the bottom of the girder and pier cap is more, suitable cribs shall be provided.
4. The girder once in position shall be suitably braced with each other so that there is no movement in them when staging, shuttering and concreting for deck slab is being done.

2.2.9.9. Staging for Casting of Deck

After placing the required number of girders between two piers, reinforcement and shuttering for end diaphragm shall be done as per drawing. Concreting of required grade shall be placed in diaphragm / cross-beam.

For vibration, needle vibrator will be used. A temporary platform shall first be erected between girders to enable the workers to fix the shutting forms and scaffolds for pouring concrete for deck slab. The staging for this shall be made with pipes and working plate form will be made by challis.

1. For Cantilever Portion:
   a. The outer girders shall be provided with hole in the webs in two rows to insert tie rods for fixing cantilever frame to support centring and shuttering form work for cantilever portion of deck slab.

   b. The frame will be fixed with girder with the help of tie rods threaded at ends and bolts over which required form work will be placed.

2. For Inner Portion:
   a. Staging for deck slab shall be supported from girder flanges and not from ground.

   b. First of all location of wooden block will be masked on bottom flange of the girder.

   c. These wooden blocks shall be fixed by 16 mm dia bolts with 200mm x 200mx10mm plate which is welded with lower pipe of H frame and Wooden blocks shall work as footing for H frame by making them with suitable slope to properly sit on bottom flange of girder.

   d. The frame shall then be lifted by crane and placed in previously marked position on bottom flange of girder.

   e. Screw Jacks shall be fixed on top of frame to carry a channel ISMC 150.

   f. To prevent sliding of the wooden blocks suitable bracing of scaffolding pipe shall be provided.

   g. The bottom shuttering of the diaphragm shall be placed in position with cut-out for the Bearings.

   h. If the gap between the shuttering and the pier cap is very small and may cause obstruction in its removal, plywood of suitable thickness shall be used as packing. Thermocoal packing shall be provided under the ply. This will enable the removal of ply board during de-shuttering.

   i. The sides of the shuttering for the diaphragm shall be placed in position only after the reinforcement has been placed in position the joints between adjacent shuttering plates shall have 5mm thick foam between them to avoid leakage of cement slurry.

   j. The top level shall be checked to match with the approved drawings. The levels shall be adjusted where required with the help of screw jacks provided over the H-frames.

   k. The shuttering surface shall be cleaned and suitable shuttering agent used for easy removal of shuttering.

l. Entire shuttering so fixed shall be duly checked and approved by AE.

2.2.9.10. Reinforcement

1. A bar bending schedule shall be prepared and shall be submitted from the Engineer's approval and the same shall be reviewed by AE if required.

2. Cutting and bending shall be done and checked for its shape and size before framing the cage.

3. The reinforcement bar shall be hoisted on to the shuttering and placed in position as indicated in the drawings.

4. Inserts for rainwater pipes and drainage spouts shall be placed in position. It will be ensured that during concreting these does not get disturbed.

5. Suitable cover blocks shall be used to maintain the required cover as shown in the drawing. Wherever necessary suitable chair shall be provided so that location of the bars does not get disturbed due to movement of labour and equipment.

6. Wooden block shall be fixed in position to provide recess for fixing of expansion joints as per approved drawing of the manufacturer.

2.2.9.11. Concreting

1. Concreting of the grade as shown in the drawings and as per the approved mix design shall be used.

2. A slump as required shall be used to achieve the required compaction.

3. Concrete from the batching plant shall be transported to site by transit mixers and pumped at required location with help of concrete pump / placer boom or crane and bucket arrangement.

4. The concrete in the diaphragm shall be laid in the layer, each layer not exceeding 500mm.

5. Concrete shall be started from one expansion joint and carried progressively to the other expansion joint.

6. The concrete pump pipe shall be so laid that by removing each length the Concrete reaches the required location and addition of pipe are not required.

2.2.9.12. De-shuttering of Deck Slab

Deck slab shuttering will be removed after concrete has achieved 14 days (as per MORTH specifications) of designed strength. The shuttering plates along with scaffolding pipe will be removed manually with the help of chain pulleys/, Manila ropes with pulleys. A temporary platform used to erection shall be useful for de-shuttering also. Shuttering plates along with scaffolding rods will be removed and kept on platform. Later these plates and scaffolding rods will be brought down by chain pulleys / Manila ropes with pulleys.

2.2.9.13. Deployment of machinery

1. Concrete pump
2. Batching plant
3. Transit mixers
4. Vibrators
5. Needles
6. Cube Mould
7. Bar Bending Machine
8. Power Arrangement (Generator/ Electricity)

2.2.9.14. Method Statement for Installation of Strip Seal type Expansion Joints with Sinus Plate

1. Expansion joints shall be installed under close supervision of the manufacture's / supplier's engineer in order to ensure the quality of the installation and its function as intended during the entire life span.

2. Detailed Installation Manual shall be supplied by the Manufacture / supplier.

3. The gap dimensions shall be pre-set depending upon the temperature at which joint is to be installed.

4. The dimensions of the recess, levels, skew & slope in the deck slab shall be established in accordance with the drawing or design data as provided by the approved manufacturer.

5. The surfaces of recess shall be thoroughly cleaned.

6. The exposed reinforcement shall be suitably adjusted to permit unobstructed lowering of the joint into the recess.

7. The recess has to the shuttered in such a way that the dimensions in joint drawing are maintained. The formwork shall be tight.

8. Immediately prior to placing the joint the pre-setting shall be inspected.

9. The lowering of joint shall be done in a pre-determined position.

10. After placement, the joint shall be levelled and finally aligned and the anchor loops on one side of the joint welded to the exposed reinforcement bars of the structure.

11. Upon completion, the same procedure shall be follow for the other side of the joint.

12. After the above procedures for every joint, the concreting procedure shall follow the clause 2610.1 (ix) of MORTH specifications.

13. Proper compaction and curing shall be done.

14. After the concrete has cured, the moveable installation brackets still in place shall be removed.

15. Rolled up neoprene strip seal shall be cut in to the required length and inserted between the edge beams by using by using a crow bar pushing the bulb of the steel in to the grooves of the edge beam. Then the sinus plate shall be fixed in position as per details shown in drawings. The carriage way surfacing shall be finished flush with the top of the steel sections, the actual junction of the surfacing wearing coat with the steal edge section shall be formed by a wedge shaped joint with a sealing compound.

2.2.9.15. Drainage Spouts

1. General:
   a. Drainage along longitudinal direction shall be ensured by sufficient numbers of drainages spouts embedded in the deck slab.

   b. The drainage spouts shall be made by corrosive resistant material such as galvanized steel and shall have a minimum diameter of 100 mm.

   c. The spacing of drainage spouts shall not exceed 10 m.

   d. The discharge from drainage spout shall be kept away from the deck structure.

2. Fabrication:
   a. All material shall be corrosion resistant. Steel components shall be of mild steel conforming to IS: 226.

   b. The drainage assembly shall be seam welded for water tightness and then hot-dip galvanized.

3. Placement:
   a. The galvanized assembly shall be given two coats of bituminous painting before placement.

   b. The whole assembly shall be placed in true position, line and level as shown in the drawings with necessary cut-out in shuttering for deck slab and held in place firmly.

   c. Where the reinforcement of the deck is required to be cut, an equivalent reinforcement shall be placed at the corners of the assembly.

   d. After setting of the deck slab concrete the shrinkage cracks around the assembly shall be totally sealed with bituminous sealant.

   e. Excess sealant should be trimmed to receive the wearing coat.

2.2.10. Methodology of Wearing Coat

A layer of BC 50 mm thick laid in single layer as per Clause 2702.1 (Type-1) of MORTH specifications. The cross slope in the deck shall be kept as per approved drawings. For the structures with flat deck surface, camber / super-elevation in the wearing coat shall be achieved by providing profile making course before laying wearing coat.

The profile making course shall be of the same material as of wearing coat. The thickness of wearing coat at any point shall not be less than that given Clause 2702.1 (Type-1) of MORTH specifications.

2.2.11. Methodology for Approach Slab

A reinforced concrete slab covering the entire width of roadway shall be provided as approach slab.

1. Minimum length of approach slab shall be kept 3.50 m or as per approved drawings.
2. Minimum thickness of slab shall be as per approved drawings.
3. The details of slab shall be according to approved drawing.
4. The cement concrete and reinforcement shall conform to sections 1600 & 1700 of MORTH specifications.
5. The base of approach slab shall be as shown in the drawing.

2.2.12. Methodology Floor Protection

2.2.12.1. Apron

1. The surface for apron laying shall be levelled and prepared for length & width as shown on drawing.
2. The size of stone should conform to clause 5.3.7.2 of IRC: 89.
3. The specific gravity of stones shall not be les then 2.65
4. The size of stone shall be such that the weight of any fragments of stone as not less than 40 kg.
5. The stone shall be hand packed within these walls.

2.2.12.2. Curtain Walls

1. First of all foundation trenches are excavated providing working space as per dimension provided in drawings for curtain walls.
2. The minimum depth of curtain wall below floor level on upstream side shall be 2.0 m and on downstream side shall be 2.5 m.
3. The curtain wall shall be in PCC of M15 grade.

2.2.13. Methodology for Backfilling Behind Structures

1. Back filling should commence in conformity with Appendix VI of IRC: 78.
2. Back filling shall be done with approved material after concrete is fully set.
3. It should be carried out in such a way as not to cause undue thrust on any part of the structure.

4. All space between foundation masonry or concrete and the side of excavation shall be refilled to original surface in layers.

5. The thickness of layer should not exceed 150mm after compaction.

6. The compaction shall be done with the help of suitable equipment such as mechanical tamper, rammer, plate vibrator etc. after watering.

7. The filter material shall be well packed to a thickness of 300 - 600 mm with smaller size towards the soil and bigger size towards the structure.

2.2.14. Methodology on Repair/Rehabilitation of Existing Bridge Structures

The current methodology may be referred for the repair, rehabilitation and strengthening of the existing structure. It is to be used in conjunction with the provisions as specified in IRC SP: 40-1993 (Guidelines for Techniques for Strengthening and Rehabilitation of Bridges) for proper understanding and detailed description.

The broad approach on repair and rehabilitation of existing structures along the project highway is as follows:

2.2.14.1. Repair/Remedial of Concrete Crash Barriers/Parapets:

Observation:
The Crash Barrier / Parapet are well constructed and erected at the location. Some of the Crash Barriers are not finished well and are partially damaged.

Remedial Measure:

1. Surface Preparation: The surface for repair will be prepared by mechanical method like chipping, sand blasting and compressed air cleaning.

2. Minor Repair of Surface: The edges of the Crash Barrier / Parapet will be repaired using 1:4 cement mortar paste.

3. Major Repair of Surface: For the major repair purpose the existing Crash Barrier / Parapet shall be repaired using the same grade of concrete and same material as that of existing or if the extent is severe than it is demolished and reconstructed.

2.2.14.2. Restoration of rusted/ exposed reinforcement of Slab:

Observation:
Overall condition of slab is good however at some locations the cover of the slab is spalled out and reinforcement got exposed.

Remedial Measure:

1. Surface Preparation: The surface for repair will be prepared by mechanical method like chipping, sand blasting and compressed air cleaning. The concrete cover is removed till the reinforcing steel is exposed completely.

2. Repair of Surface:
   a) Firstly the rust of the reinforcement is removed by sand blasting, needle hammer and wire brushing.

b) The reinforcement is protected towards corrosion by epoxy coating or encapsulated in an alkaline coating i.e. cementitious bond coat. After protection of the reinforcement, the slab is filled by the concrete of one grade higher if sufficient thickness is available. If the sufficient thickness is not available the slab is filled with polymer cement mortar.

2.2.14.3. Repair of Foundation

It is not possible to evolve a general method of repair and/or strengthening of foundations.

1. Scour and Erosion protection and repair of washed away/damaged protection works.
2. Repair of Foundation built on soft ground subjected to erosion.
3. Repair of serrations on the surface caused by high velocity of particles in the water.

Observation:
Excessive scour is one of the more frequent factors that may cause or lead to structural failure or distress of the foundation. The degree of damage depends on the factors such as the stream bed material, the intensity of discharge, the silt charge, obliquity of stream flow and the shape of the structure.

Remedial Measure:
To restrict scour around pier it is common to utilize 'garlanding technique' i.e. placement of very heavy concrete blocks or stones around it or by sheet piling or by stone pitching

2.2.14.4. Repair of Sub-structure of Minor Bridges:

Observation:
Deterioration of joints, spalling, disintegration of wall of the abutment.

Remedial Measure:
The deteriorating joints may be repaired by injection of epoxy mortar surface protection and plastering the locations using 1:4 cement mortar paste on the entire surface.

2.2.14.5. Repair of Concrete

If there is a partial depth repair in concrete in that case, remove concrete up to 75-100mm depth and about 1m by 1m wide (depending on the type of deterioration); then fill it with elastomeric concrete/polymer concrete by applying some reinforcement in form of chicken mesh and the grade of elastomeric concrete should be one grade higher than the existing designed concrete. The reinforcement should be fitted in the existing concrete using u clips.

2.2.14.6. Repair of Cracks

Observation:
Check the crack for its life i.e. it is active or dormant.

Remedial Measure:
1. If the crack is active the repair it by stitching, jacketing, injection of epoxy resin and mortar.
2. If the crack is dormant it may be repaired by grouting, jacketing, thin resurfacing.

2.2.14.7. Repair for Drainage Spout

Repair / replacement of drainage spout to be done as per site condition.

2.2.14.8. Replacement of Approach Slab

To be decided as per site condition.

2.2.14.9. Repair / Replacement of Bearing / Expansion Joint

For bearing / expansion joint the methodology for repair / replacement shall be as per the specification of the vendor recommendations.